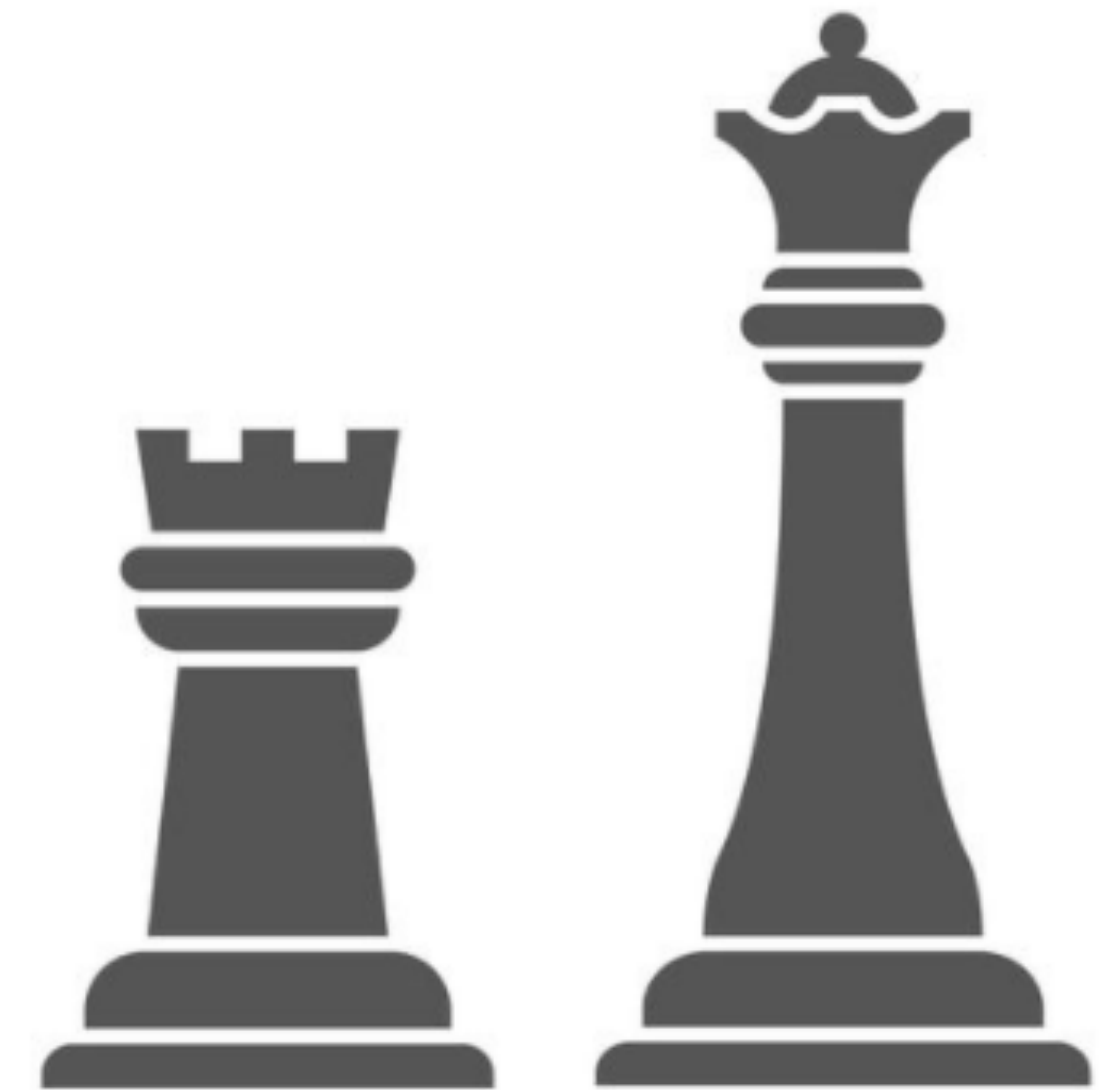


Lecture 5: Spatial weights

Instructor: Michael Szell

Feb 25, 2026



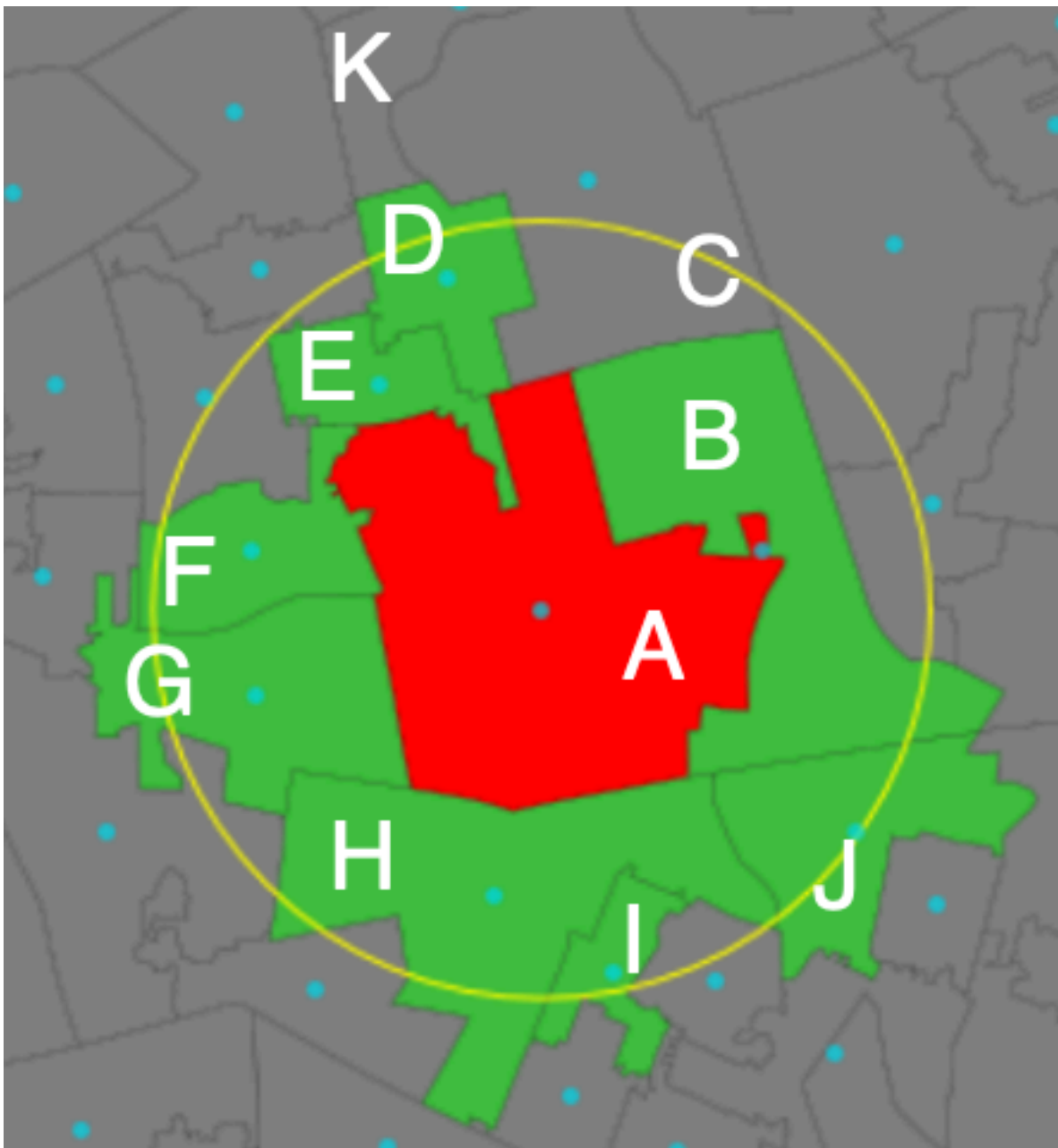
GDS asks: How do things relate in space?

Everything is related to everything else,
but near things are more related than
distant things.

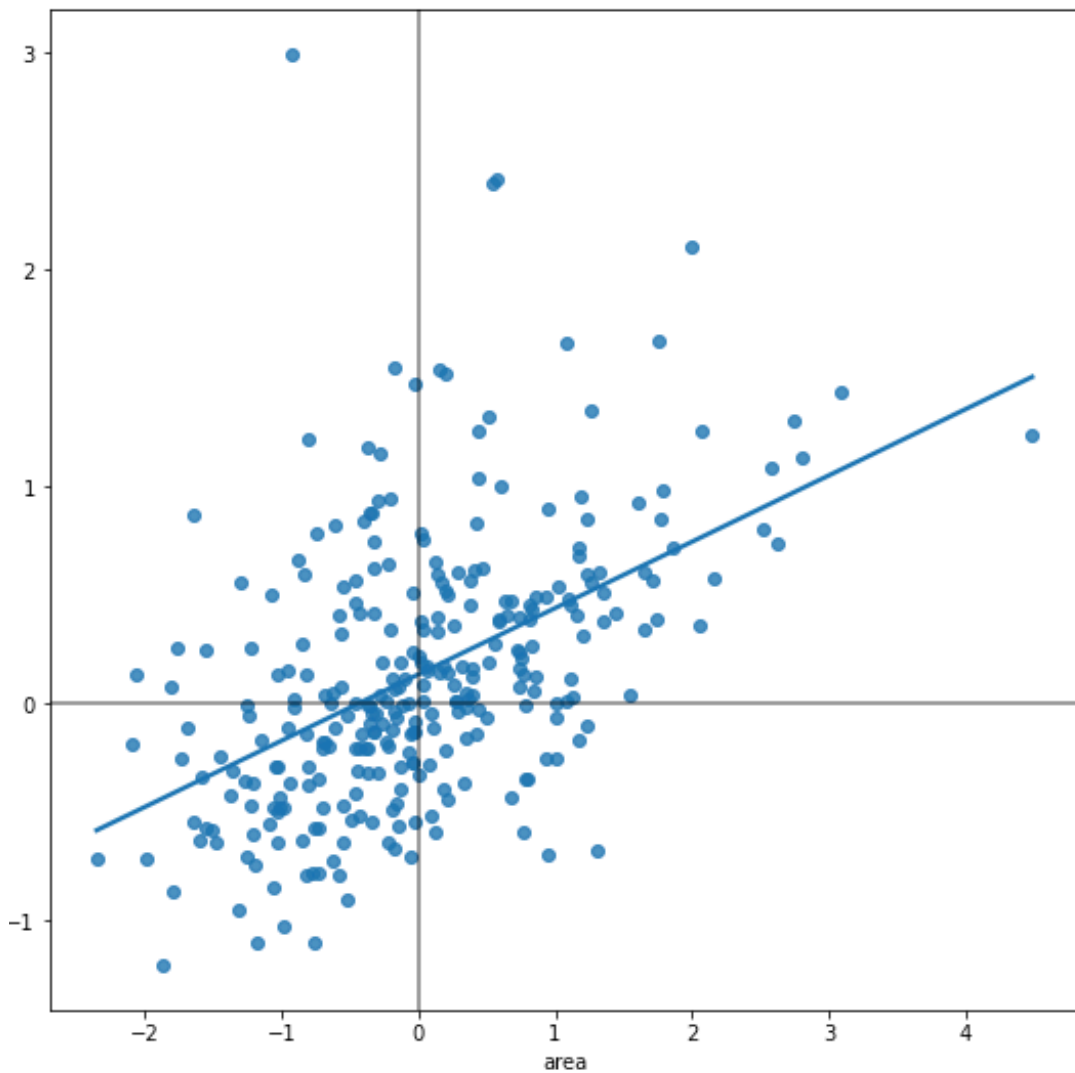
Tobler's 1st law of geography

Today you will learn about spatial weight matrices

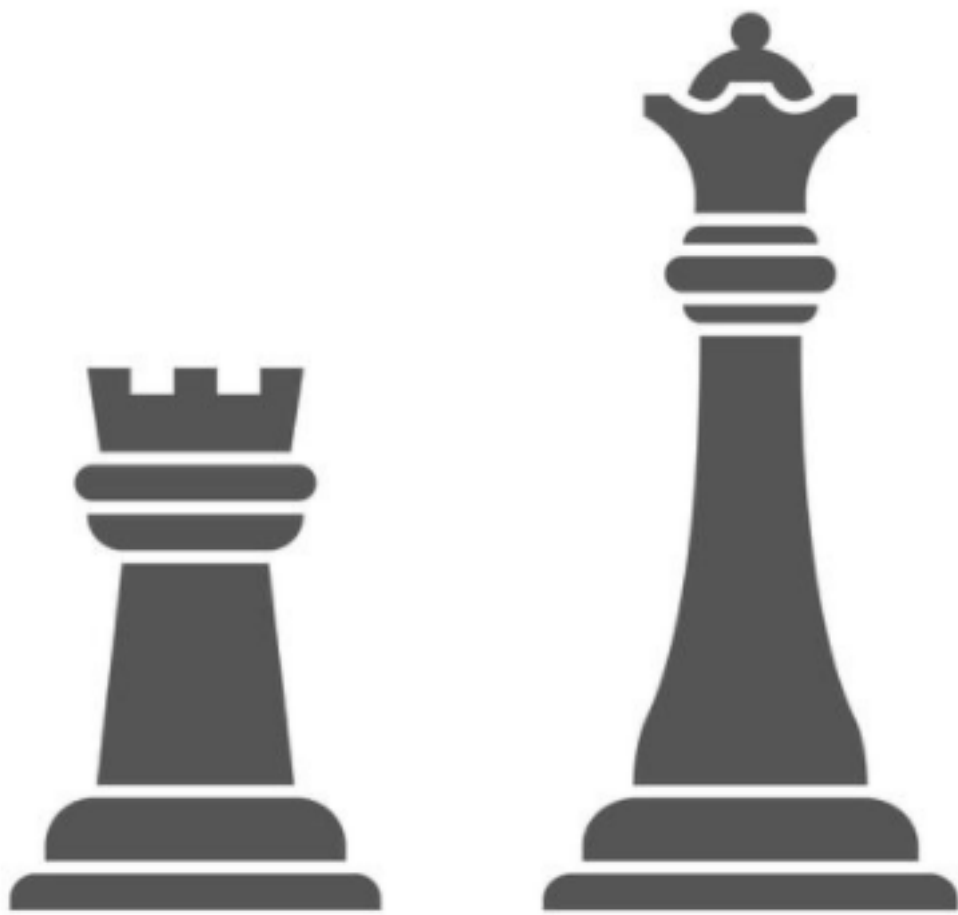
What they are,
why they are important



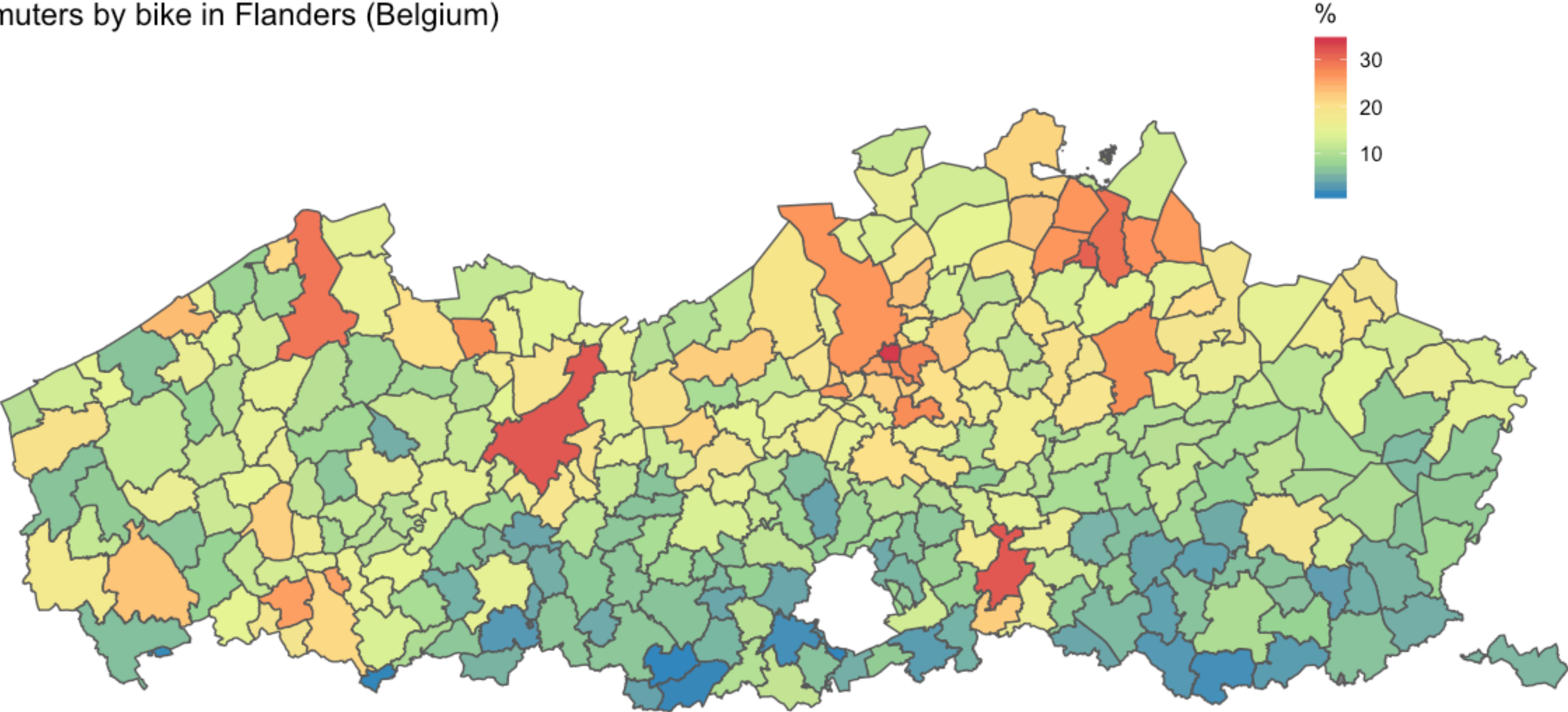
How to use them
in Python



Ways of defining them



Commuters by bike in Flanders (Belgium)



Data: Gemeentemonitor 2017

How are objects related to each other in space?

The **spatial weight matrix** W encodes the spatial relation between N objects

$$W = \begin{pmatrix} 0 & w_{12} & \dots & w_{1N} \\ w_{21} & \ddots & w_{ij} & \vdots \\ \vdots & w_{ji} & 0 & \vdots \\ w_{N1} & \dots & \dots & 0 \end{pmatrix}$$

N times N , positive

$$w_{ii} = 0$$

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N times N, positive
 $w_{ii} = 0$

Generally, all non-zero elements in a row i are called the **neighbors** of object i . How to define neighbour?

How are objects related to each other in space?

The **spatial weight matrix** W encodes the spatial relation between N objects

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N times N , positive

$$w_{ii} = 0$$

Contiguity

Is object 2 "next to" object 1?

How are objects related to each other in space?

The **spatial weight matrix** W encodes the spatial relation between N objects

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N times N, positive
 $w_{ii} = 0$

Contiguity

Is object 2 "next to" object 1?

Distance

Is object 2 "close" to object 1?

How are objects related to each other in space?

The **spatial weight matrix** W encodes the spatial relation between N objects

$$W = \begin{pmatrix} 0 & w_{12} & \dots & w_{1N} \\ w_{21} & \ddots & w_{ij} & \vdots \\ \vdots & w_{ji} & 0 & \vdots \\ w_{N1} & \dots & \dots & 0 \end{pmatrix}$$

N times N , positive
 $w_{ii} = 0$

Contiguity

Is object 2 "next to" object 1?

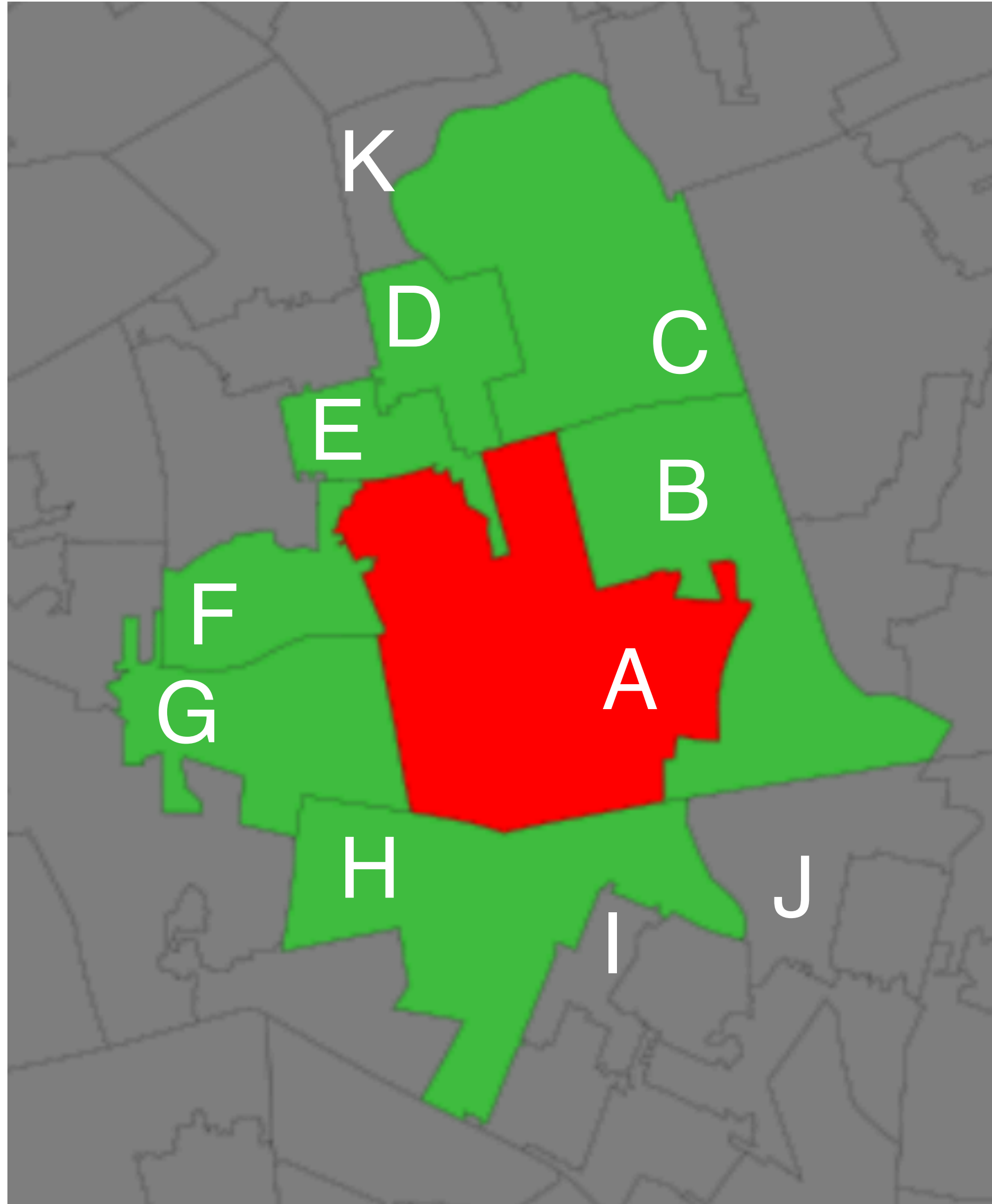
Distance

Is object 2 "close" to object 1?

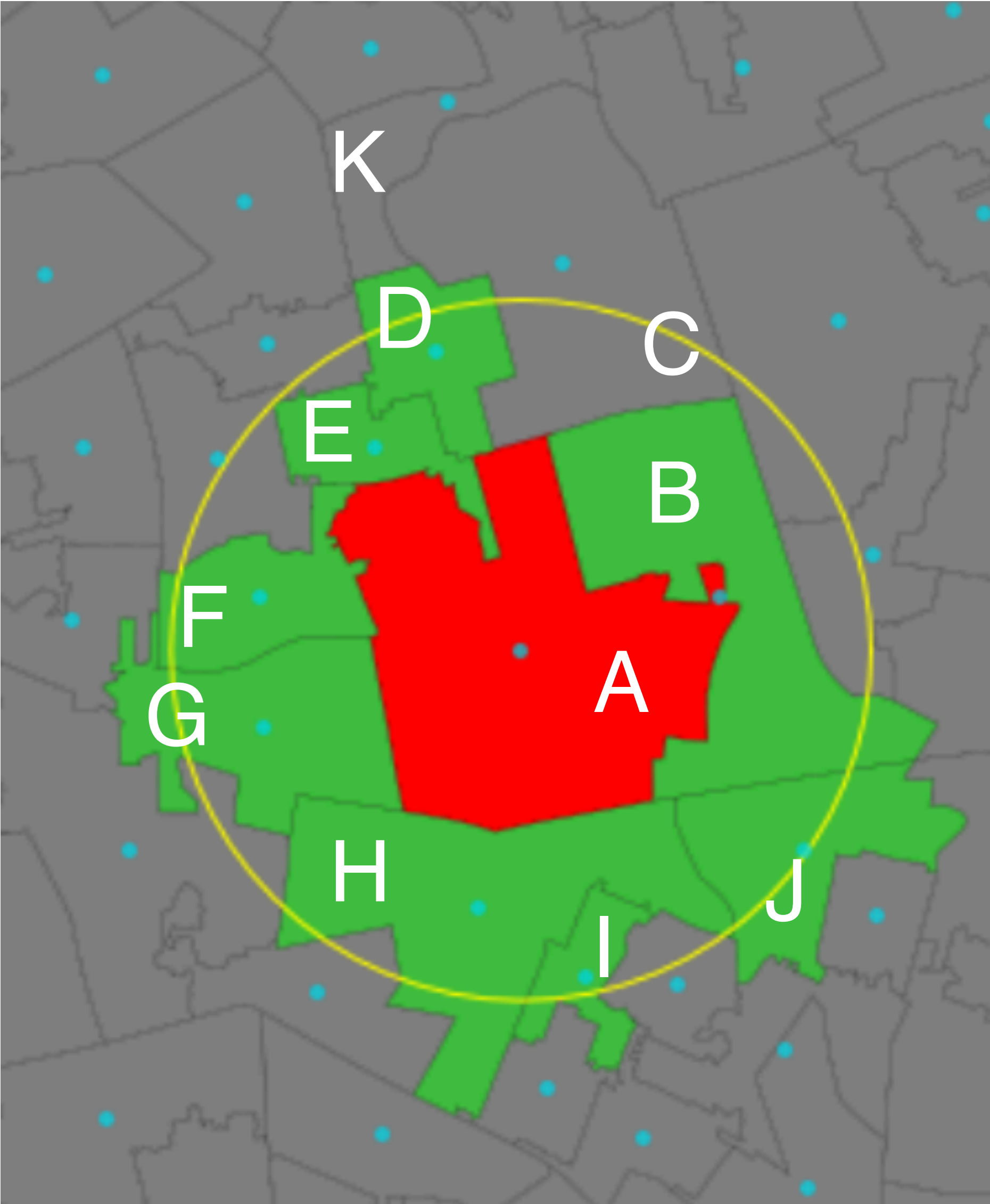
Block

Is object 2 in the same "place" as object 1?

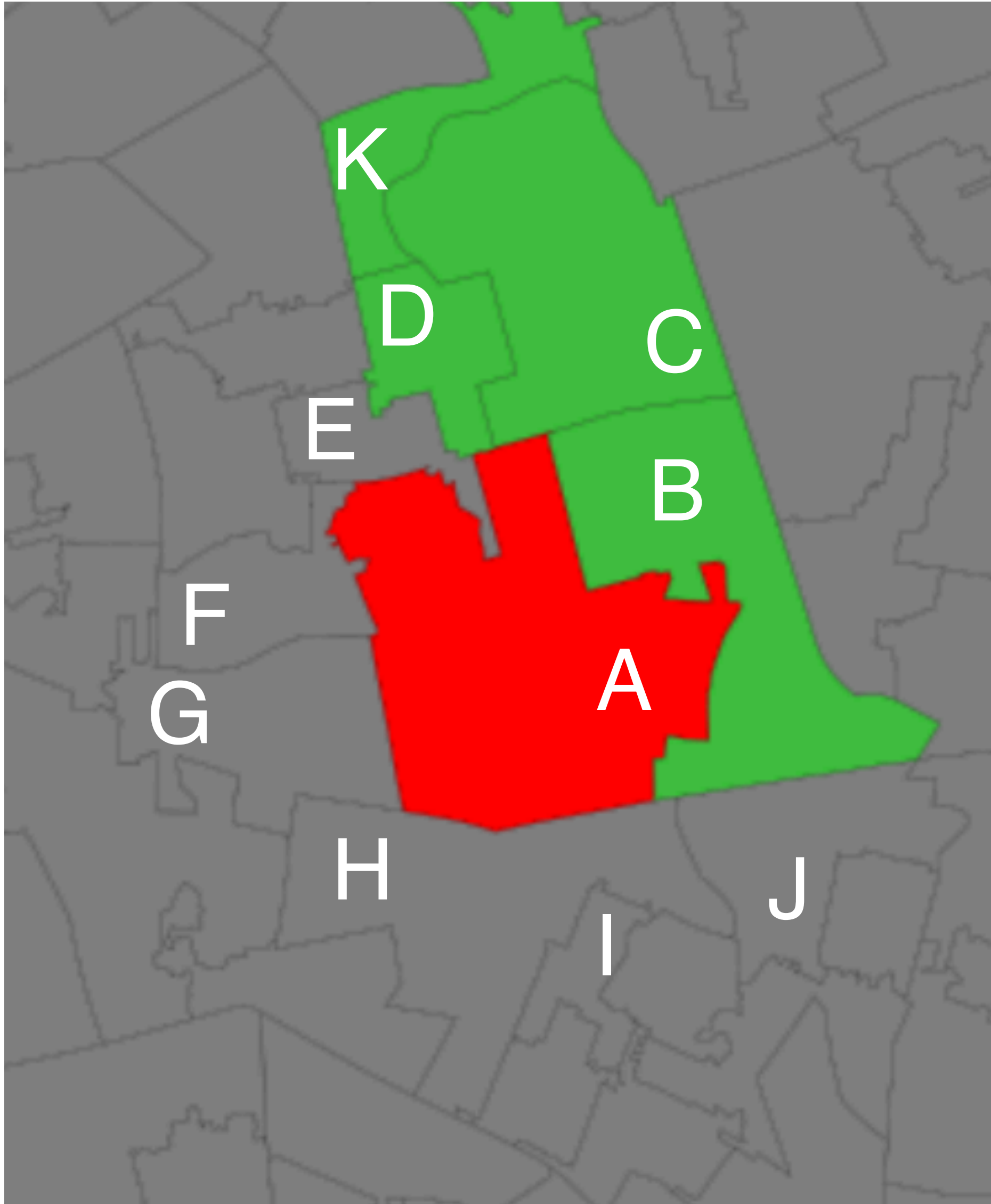
Contiguity-based W: sharing a boundary

[illegible]

Distance-based W: being closer than a threshold

[illegible]

Block-based W: being in the same administrative unit

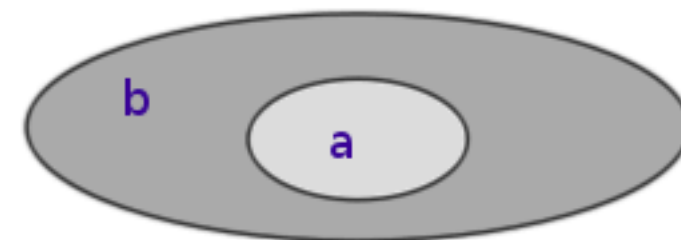
[illegible]

Special cases and variations

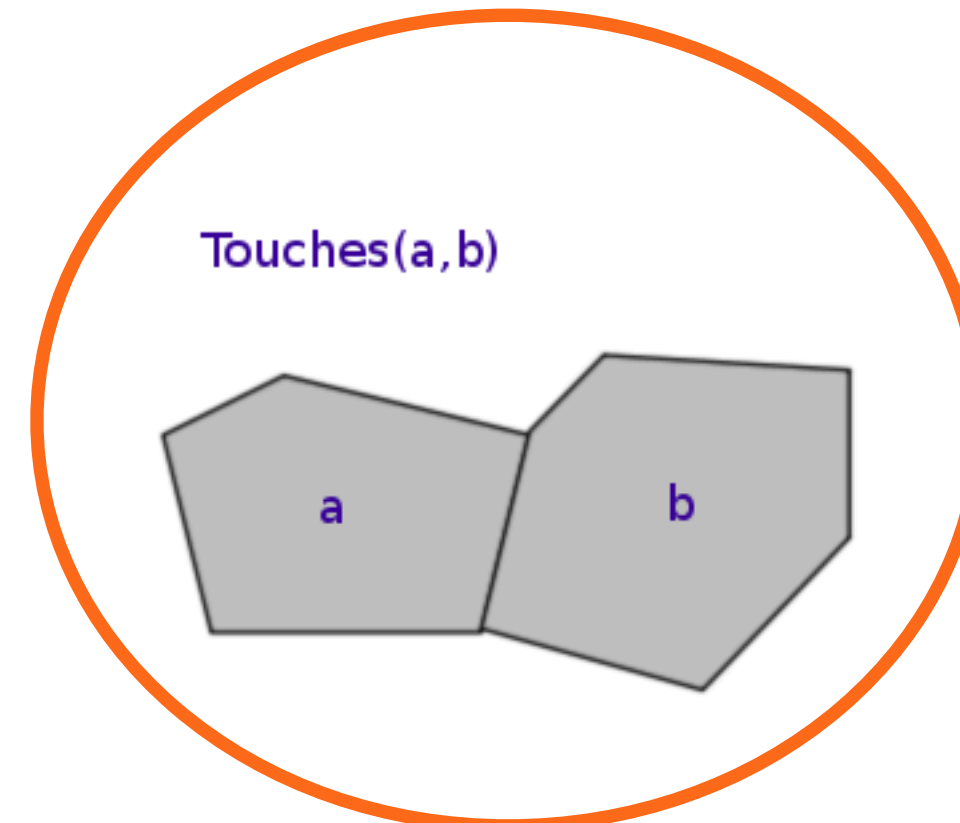
Geospatial **topology** is the study and application of qualitative spatial relationships between geographic features

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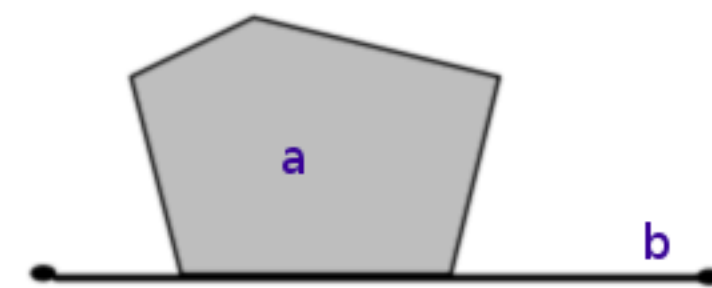
Within(a,b)



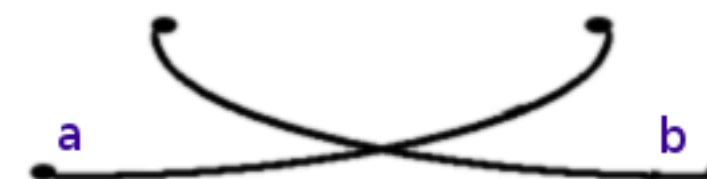
Touches(a,b)



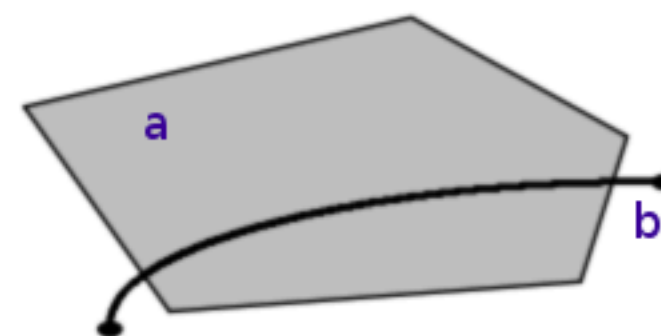
Touches(a,b)



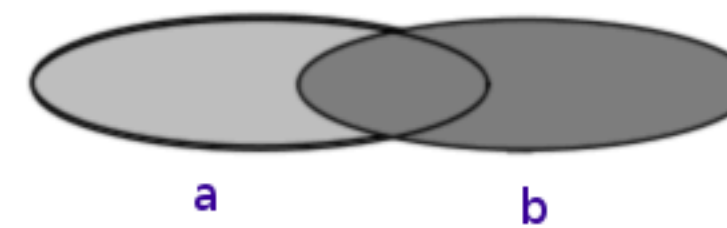
Crosses(a,b)



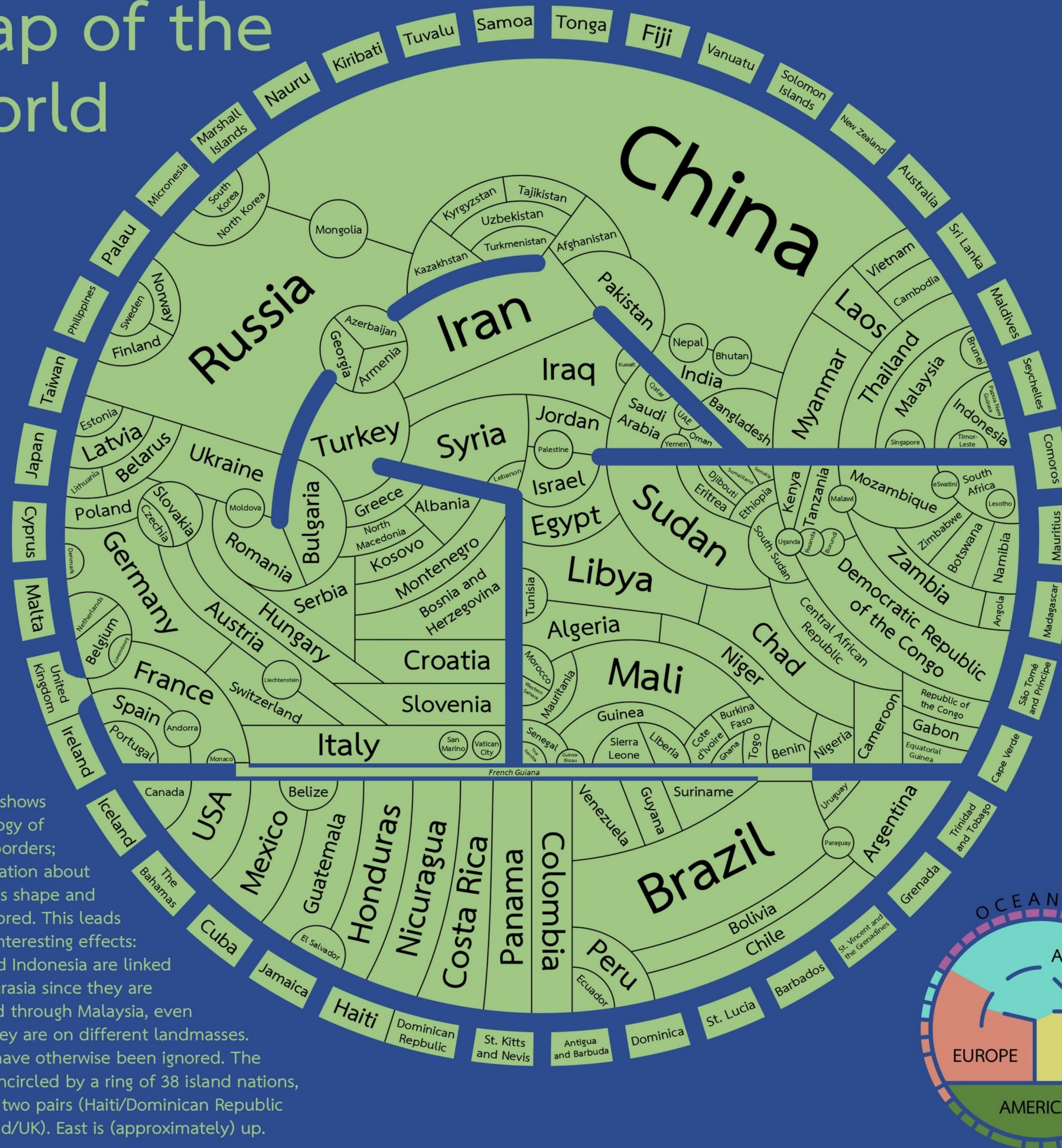
Crosses(a,b)



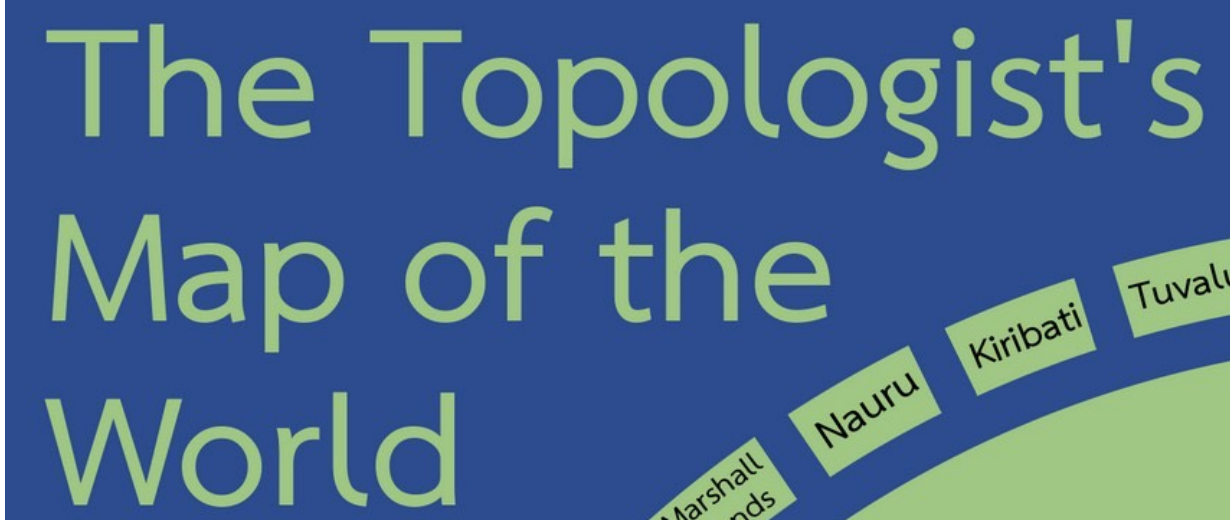
Overlaps(a,b)



The Topologist's Map of the World

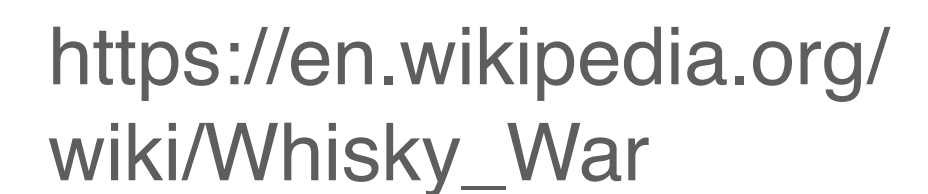
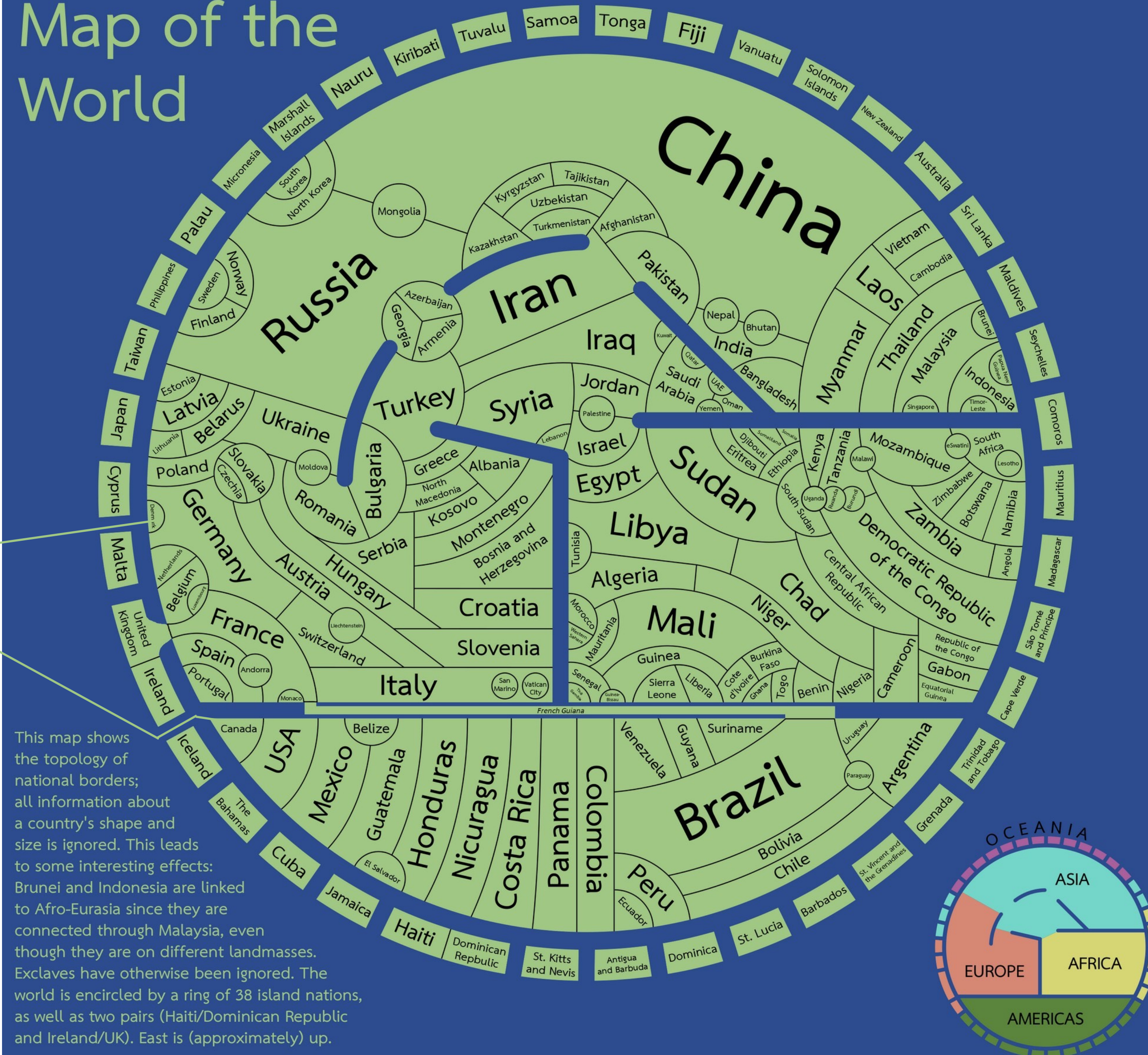


This map shows the topology of national borders; all information about a country's shape and size is ignored. This leads to some interesting effects: Brunei and Indonesia are linked to Afro-Eurasia since they are connected through Malaysia, even though they are on different landmasses. Exclaves have otherwise been ignored. The world is encircled by a ring of 38 island nations, as well as two pairs (Haiti/Dominican Republic and Ireland/UK). East is (approximately) up.

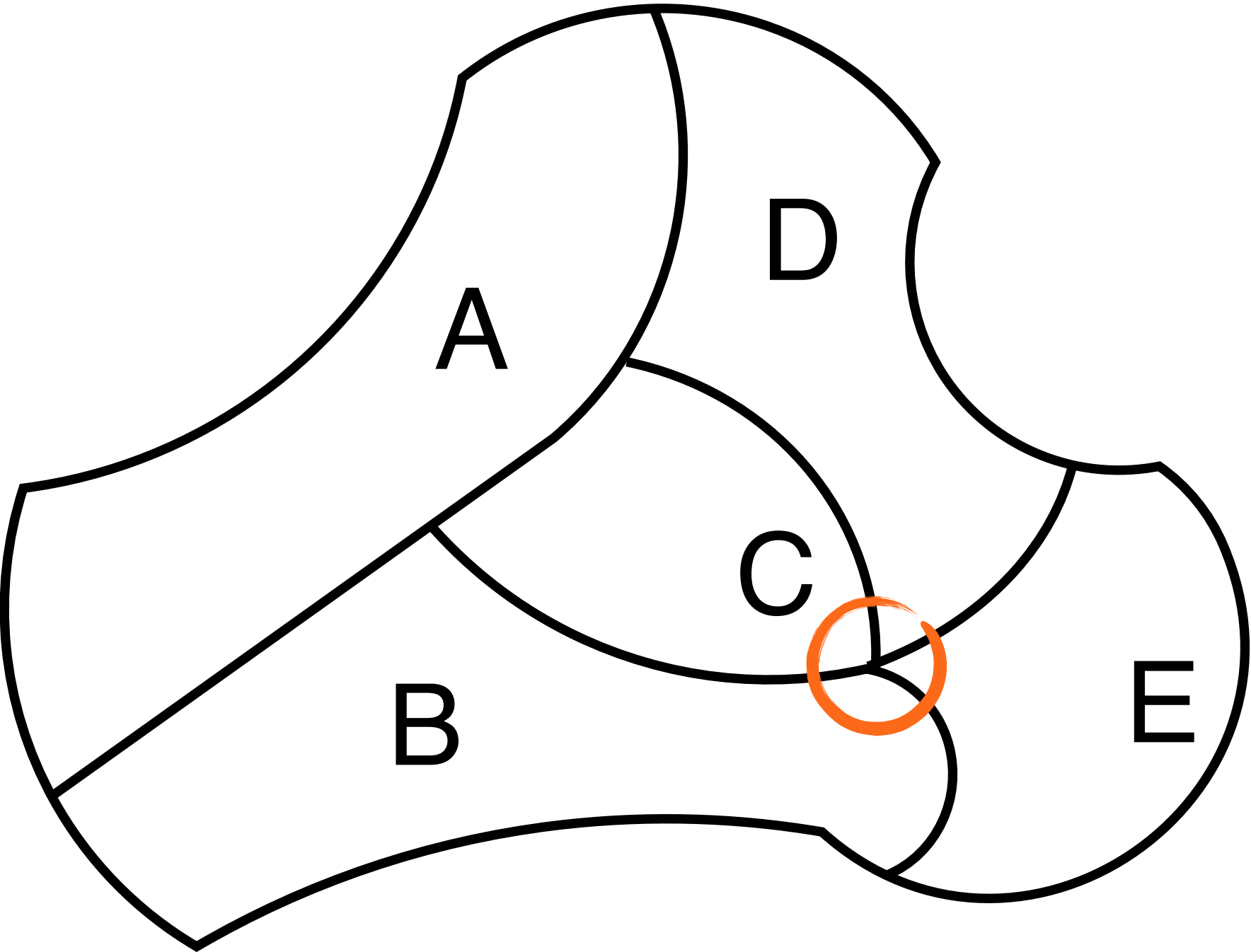


Hans island

This map shows the topology of national borders; all information about a country's shape and size is ignored. This leads to some interesting effects: Brunei and Indonesia are linked to Afro-Eurasia since they are connected through Malaysia, even though they are on different landmasses. Exclaves have otherwise been ignored. The world is encircled by a ring of 38 island nations, as well as two pairs (Haiti/Dominican Republic and Ireland/UK). East is (approximately) up.

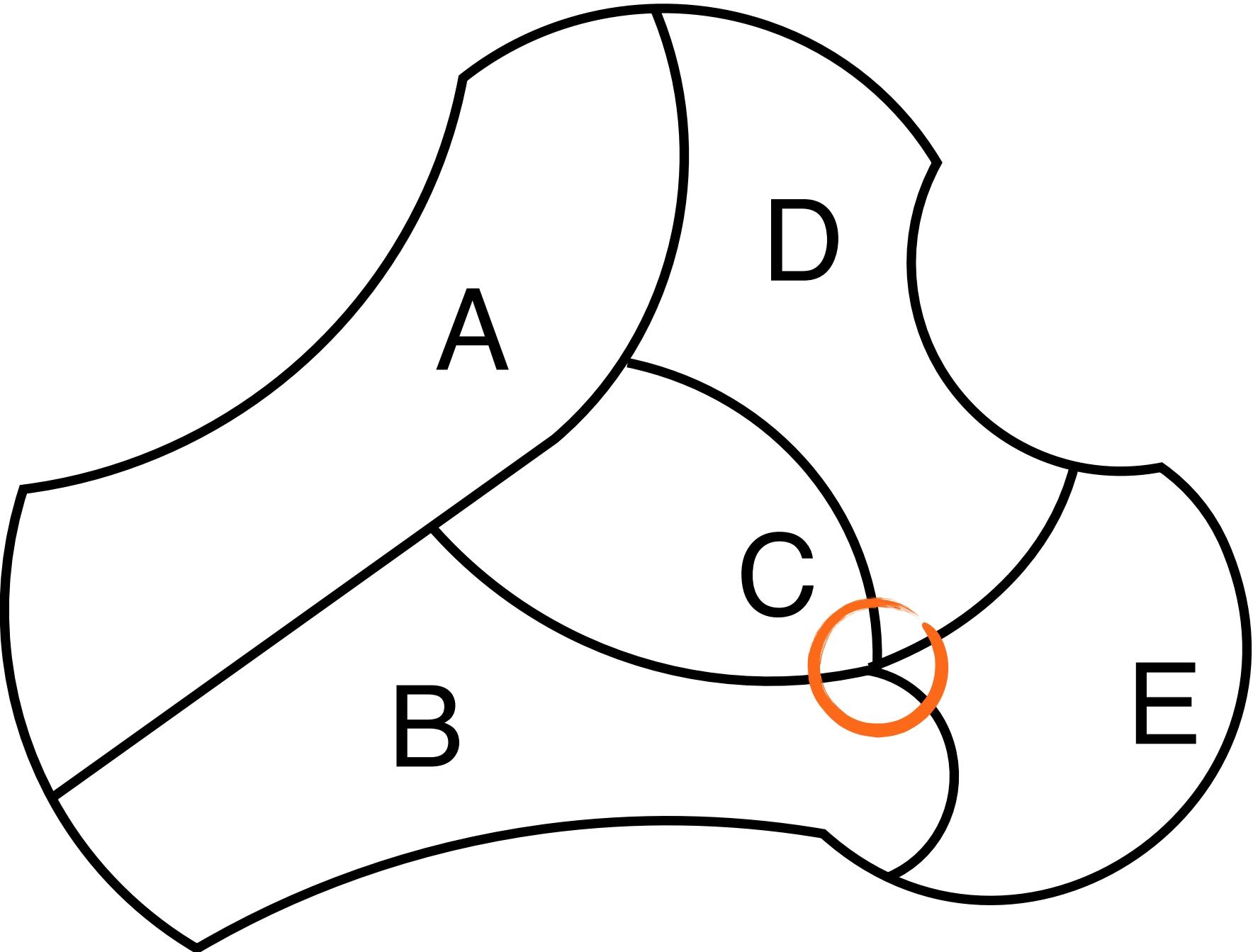


Contiguity-based W: What is a boundary?



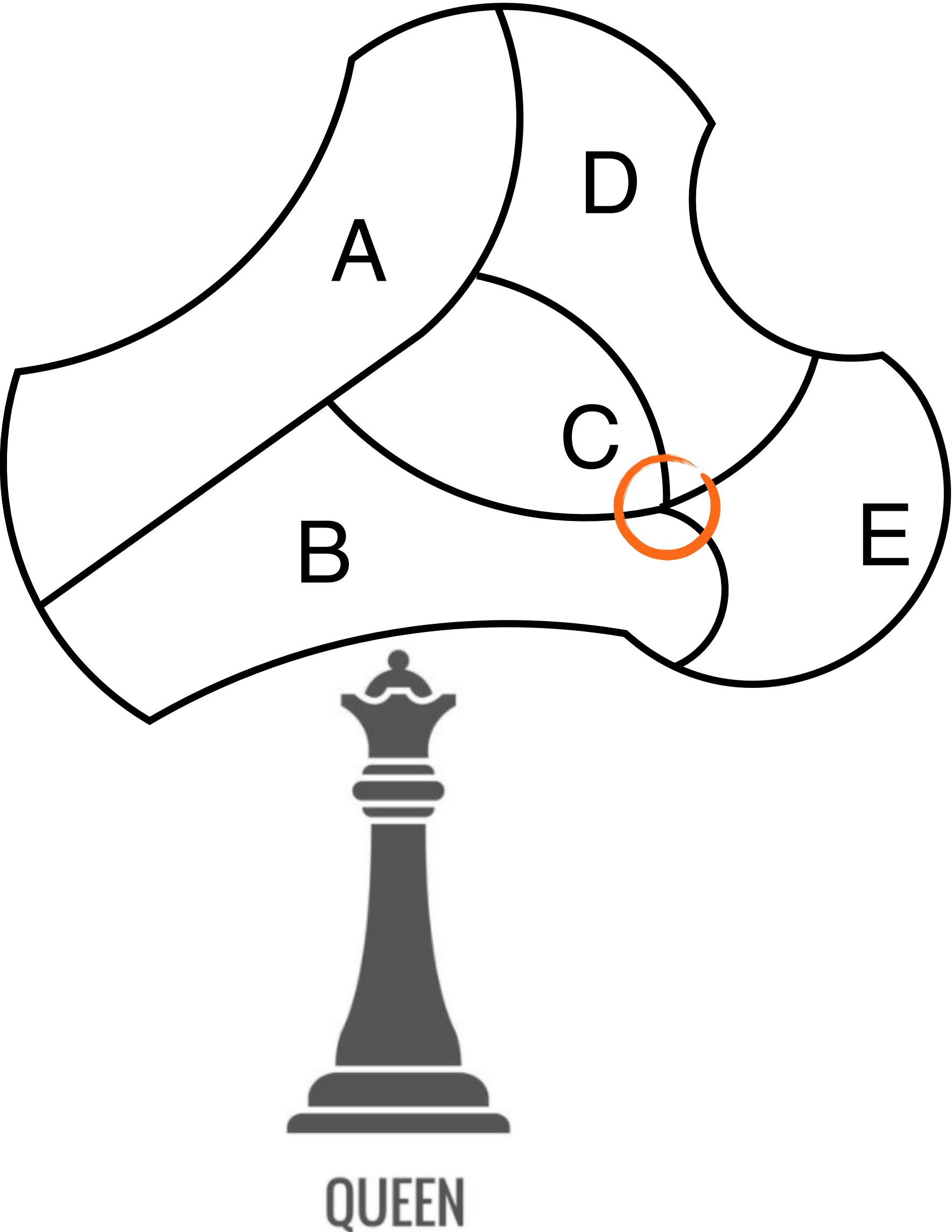
	A	B	C	D	E
A	0	1	1	1	0
B	1	0	1	?	1
C	1	1	0	1	?
D	1	?	1	0	1
E	0	1	?	1	0

Contiguity-based W: What is a boundary?



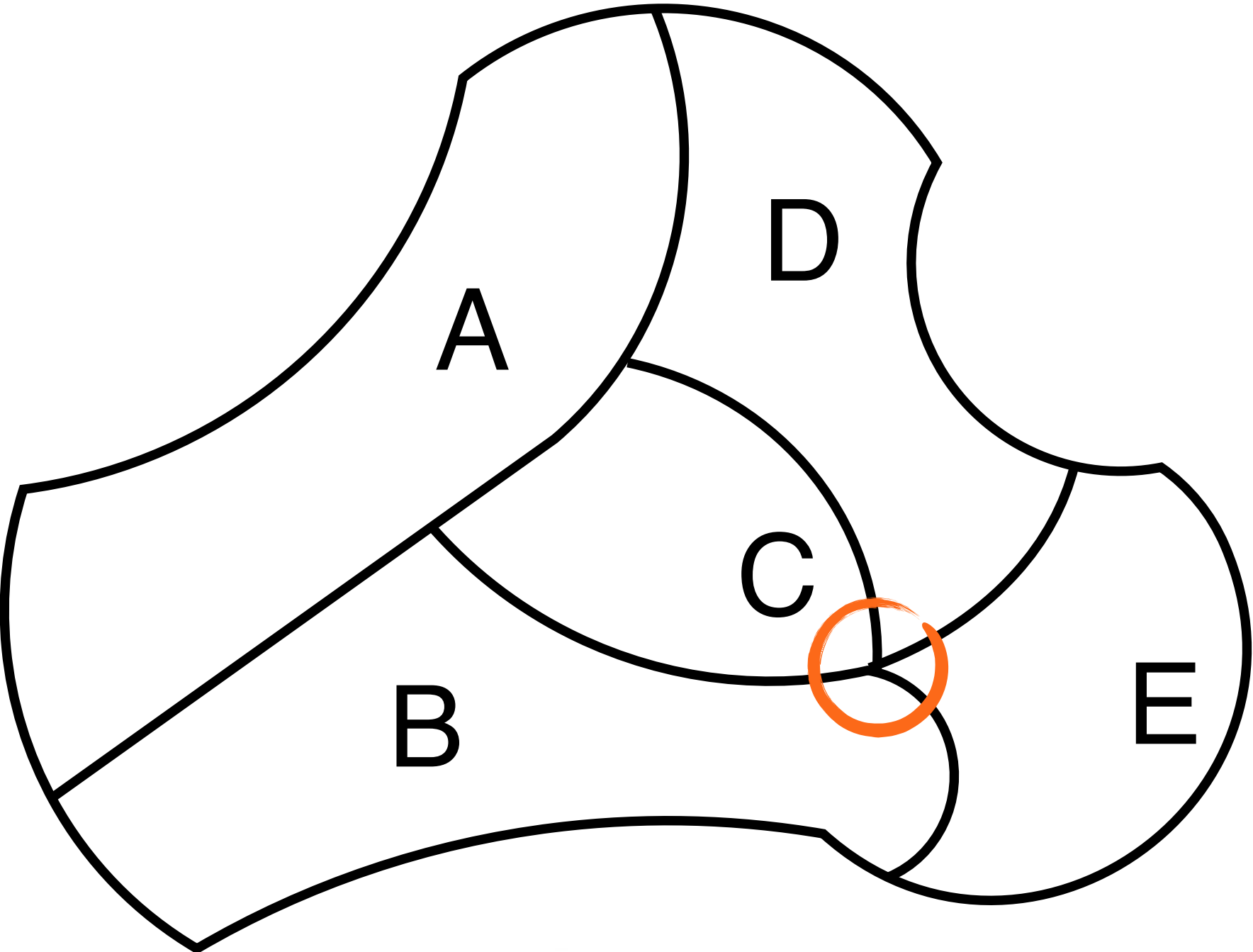
	A	B	C	D	E
A	0	1	1	1	0
B	1	0	1	0	1
C	1	1	0	1	0
D	1	0	1	0	1
E	0	1	0	1	0

Contiguity-based W: What is a boundary?



	A	B	C	D	E
A	0	1	1	1	0
B	1	0	1	1	1
C	1	1	0	1	1
D	1	1	1	0	1
E	0	1	1	1	0

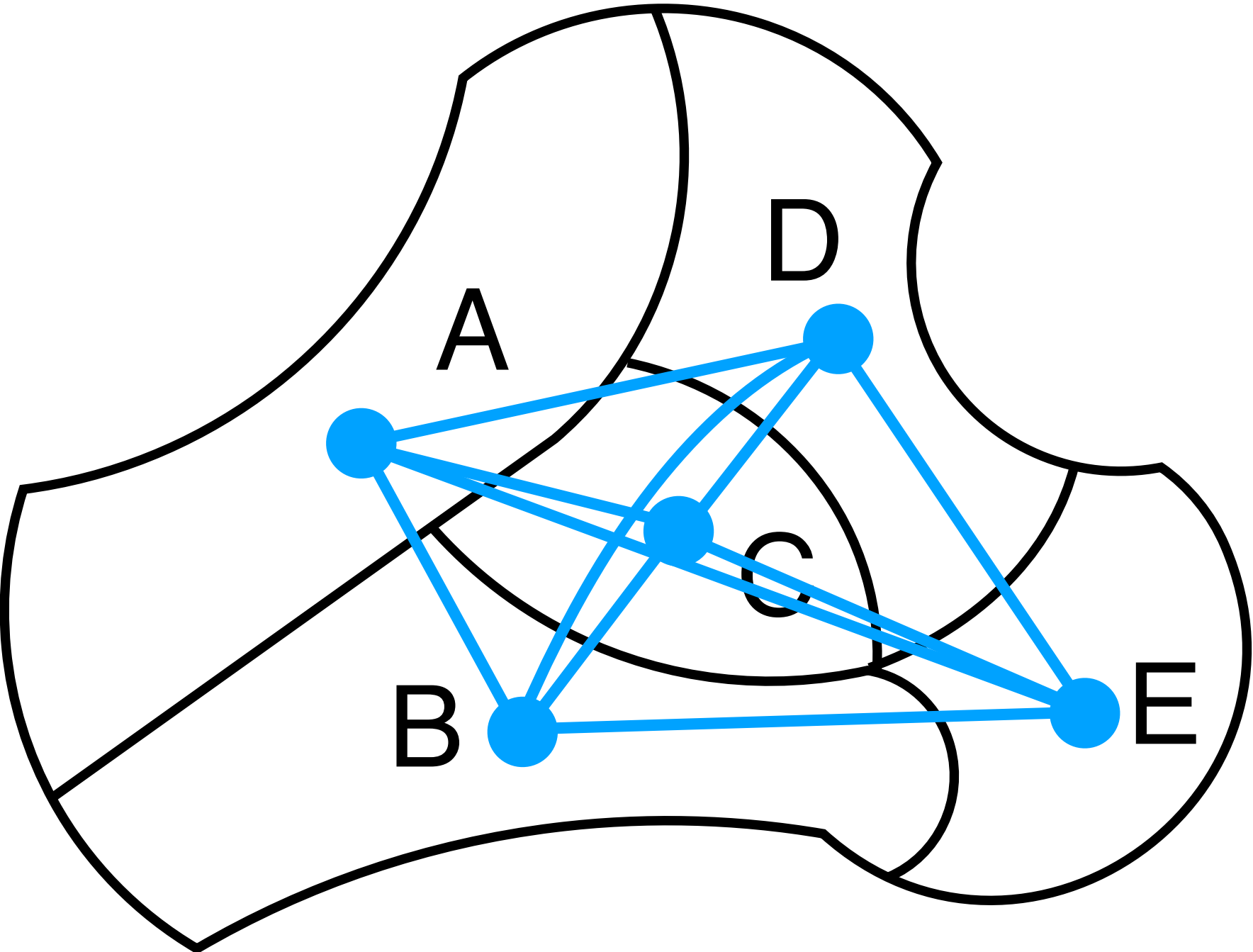
Contiguity-based W: What is a boundary?



BISHOP

	A	B	C	D	E
A	0	0	0	0	0
B	0	0	0	1	0
C	0	0	0	0	1
D	0	1	0	0	0
E	0	0	1	0	0

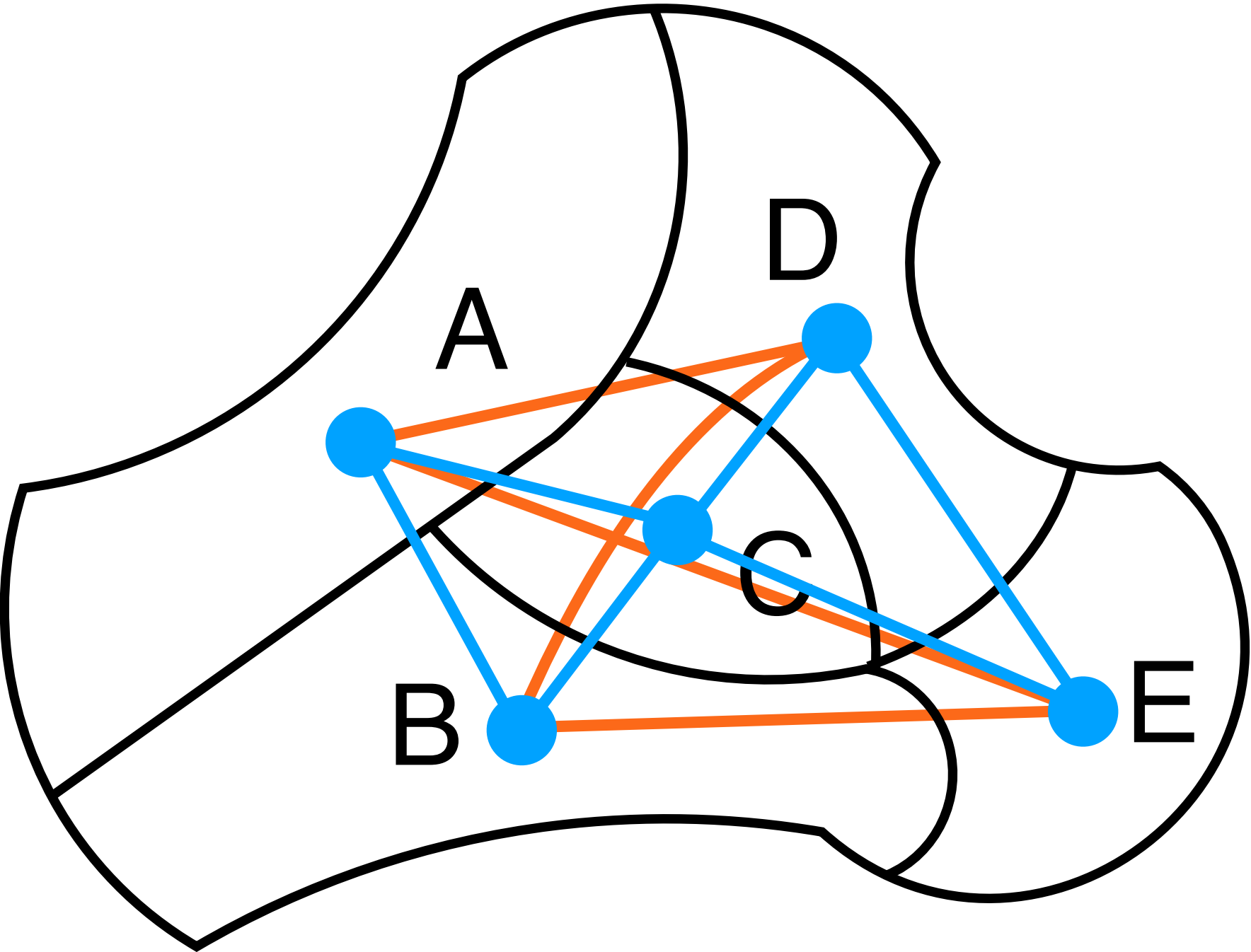
Distance-based W: Values can be continuous



$$w_{ij} = \frac{1}{d_{ij}} \quad \text{or} \quad w_{ij} = \frac{1}{d_{ij}^2}$$

	A	B	C	D	E
A	0	0.58	0.57	0.39	0.25
B	0.58	0	0.76	0.38	0.33
C	0.57	0.76	0	0.76	0.43
D	0.39	0.38	0.76	0	0.42
E	0.25	0.33	0.43	0.42	0

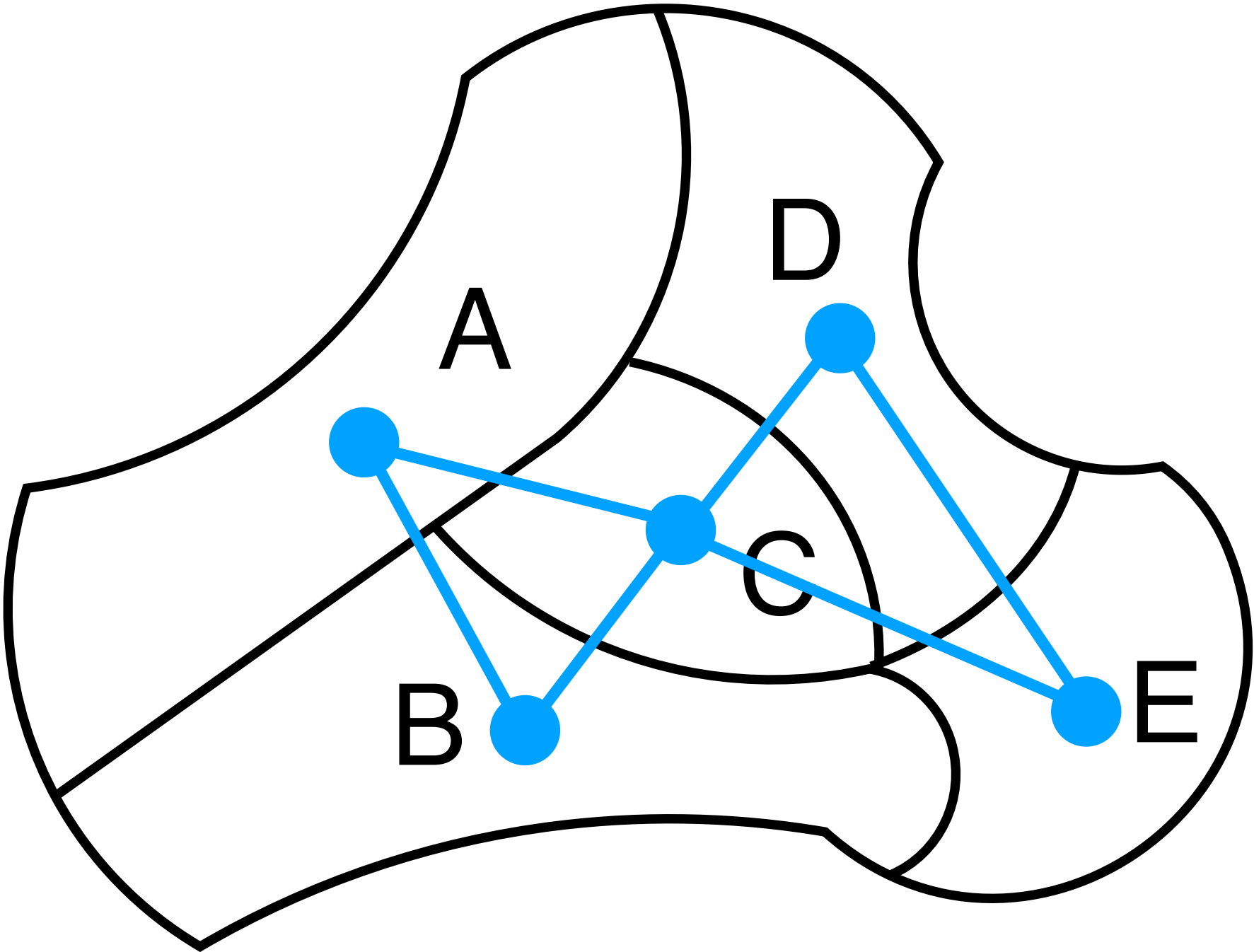
Distance-based W: Values can be continuous



We want a **threshold**
to keep W sparse!

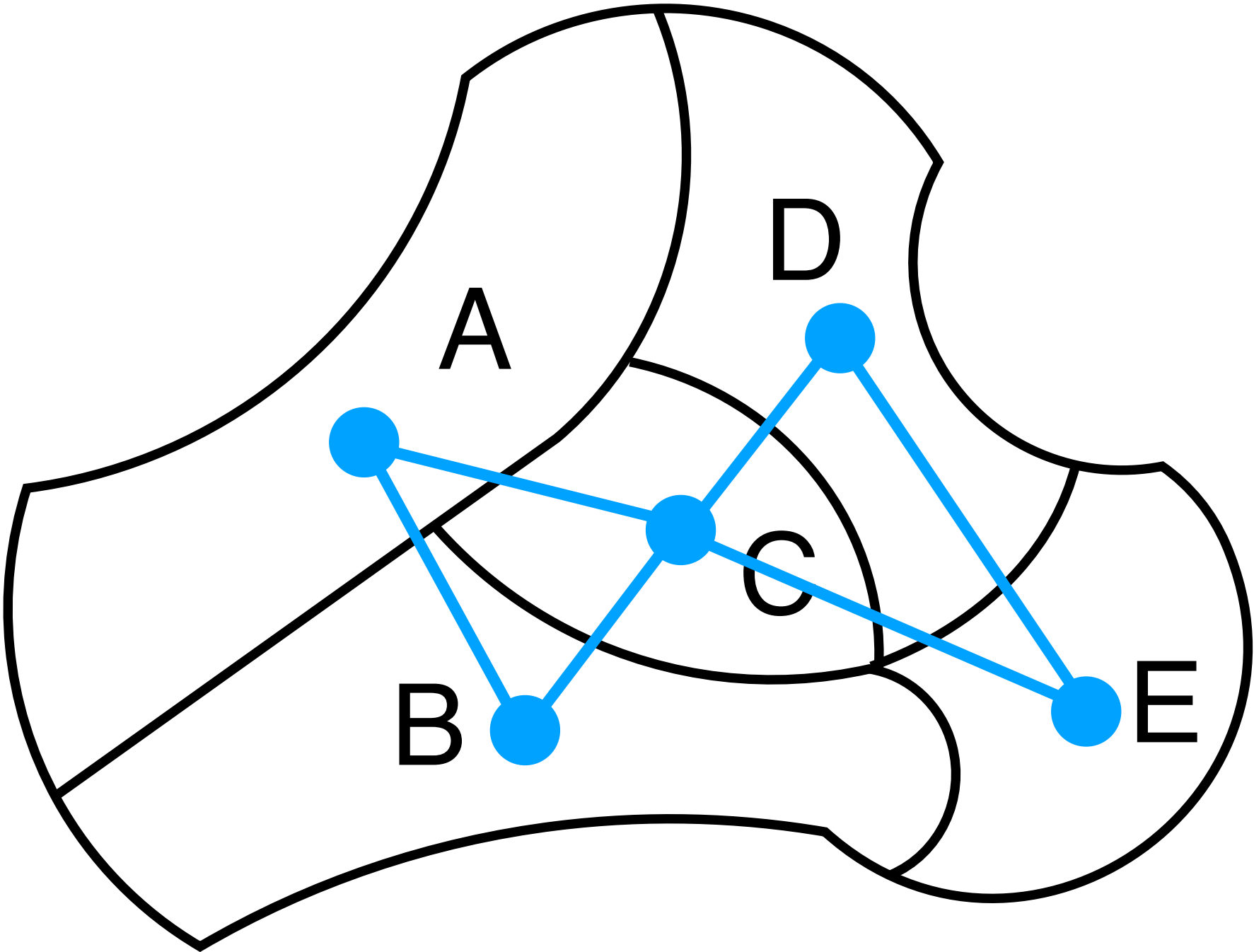
t=0.4	A	B	C	D	E
A	0	0.58	0.57	0	0
B	0.58	0	0.76	0	0
C	0.57	0.76	0	0.76	0.43
D	0	0	0.76	0	0.42
E	0	0	0.43	0.42	0

Distance-based W: KNN (k closest neighbors)



k=2	A	B	C	D	E
A	0	1	1	0	0
B	1	0	1	0	0
C	0	1	0	1	0
D	0	0	1	0	1
E	0	0	1	1	0

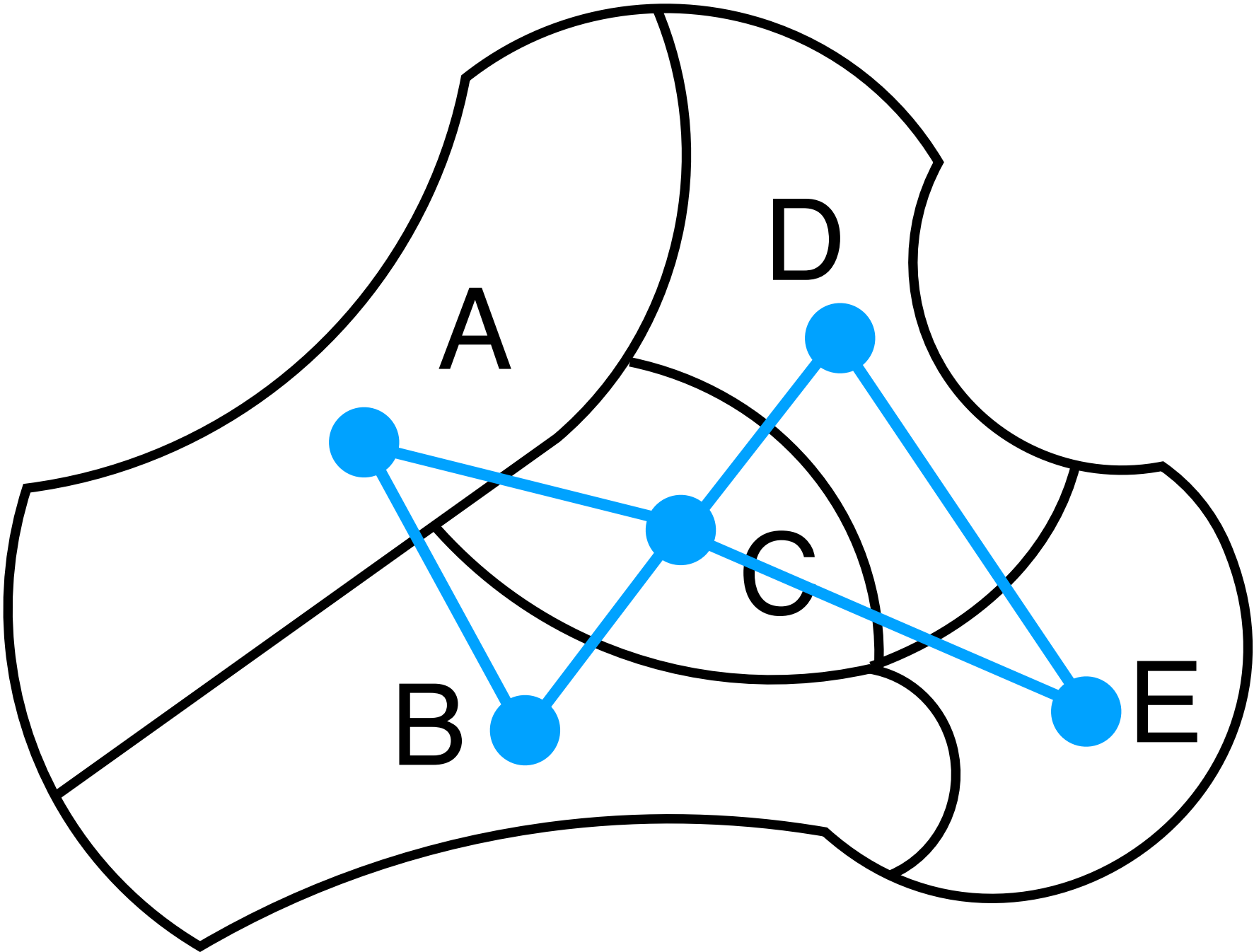
Distance-based W: KNN (k closest neighbors)



Anything weird
going on?

k=2	A	B	C	D	E
A	0	1	1	0	0
B	1	0	1	0	0
C	0	1	0	1	0
D	0	0	1	0	1
E	0	0	1	1	0

Distance-based W: KNN (k closest neighbors)

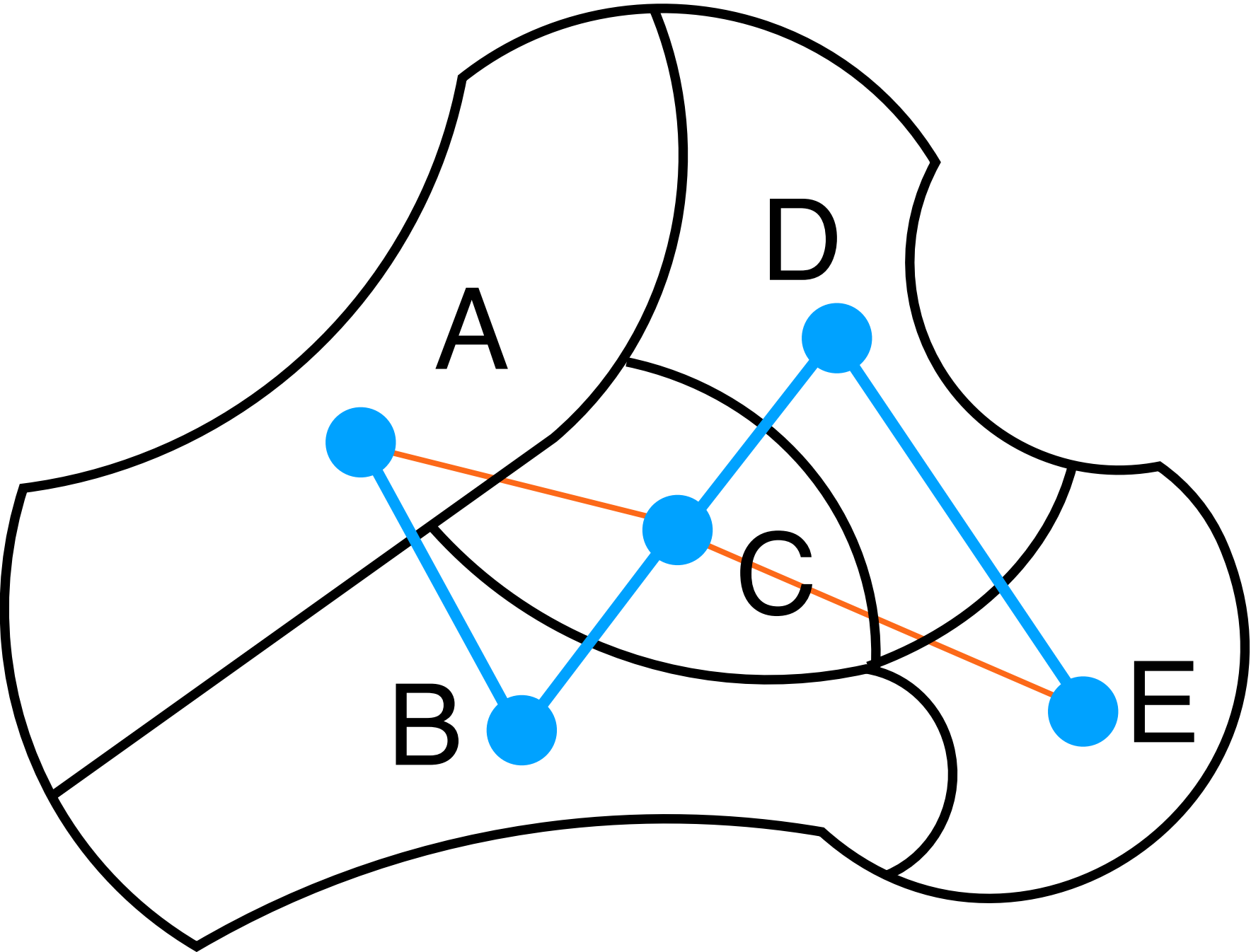


Anything weird
going on?

k=2	A	B	C	D	E
A	0 0	0.58 1	0.57 1	0 0	0 0
B	1	0	1	0	0
C	0.57 0	0.76 1	0 0	0.76 1	0.43 0
D	0	0	1	0	1
E	0	0	1	1	0

The matrix has become asymmetric

Distance-based W: KNN (k closest neighbors)

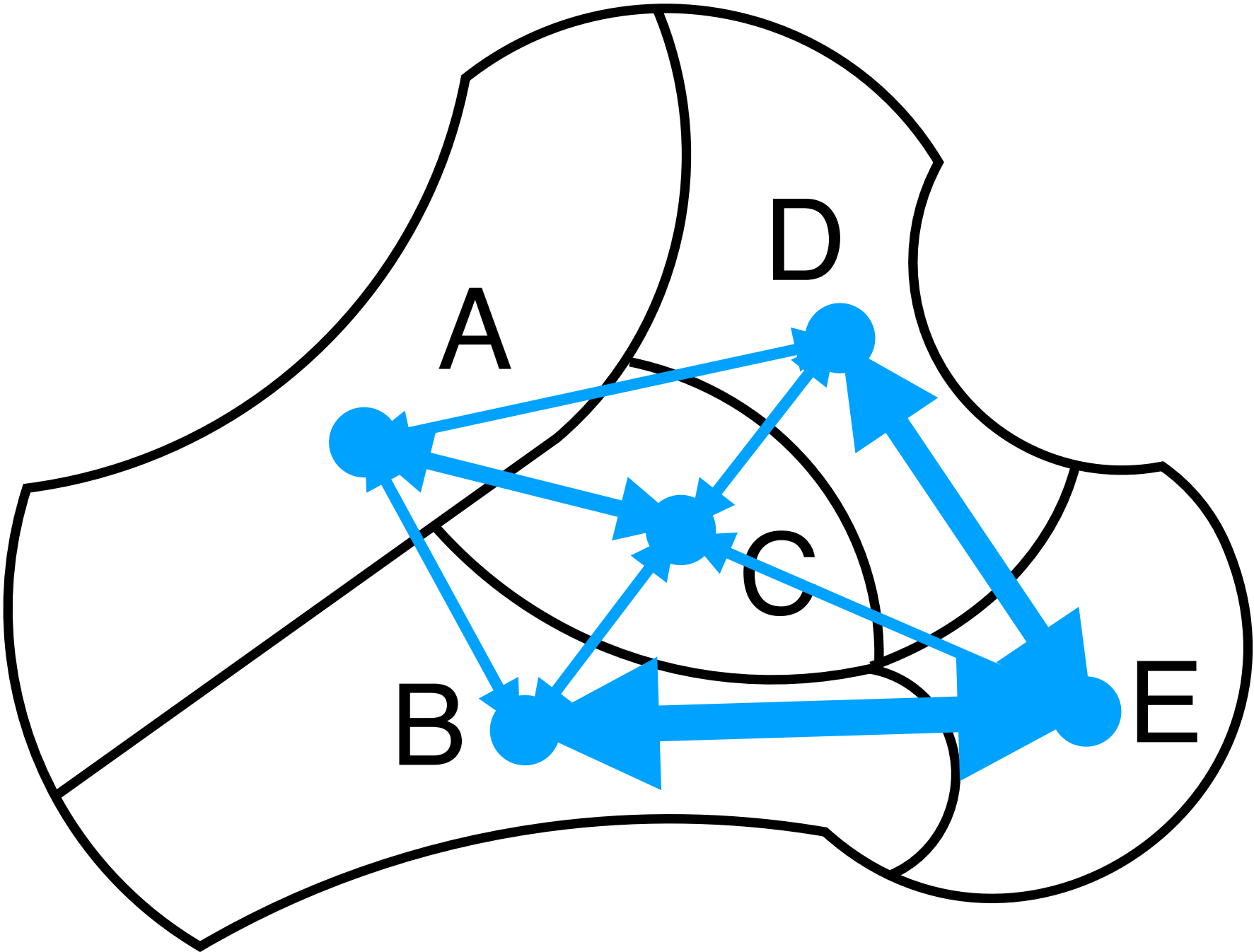


To make it symmetric, you could use:

$$(W + W^T)/2$$

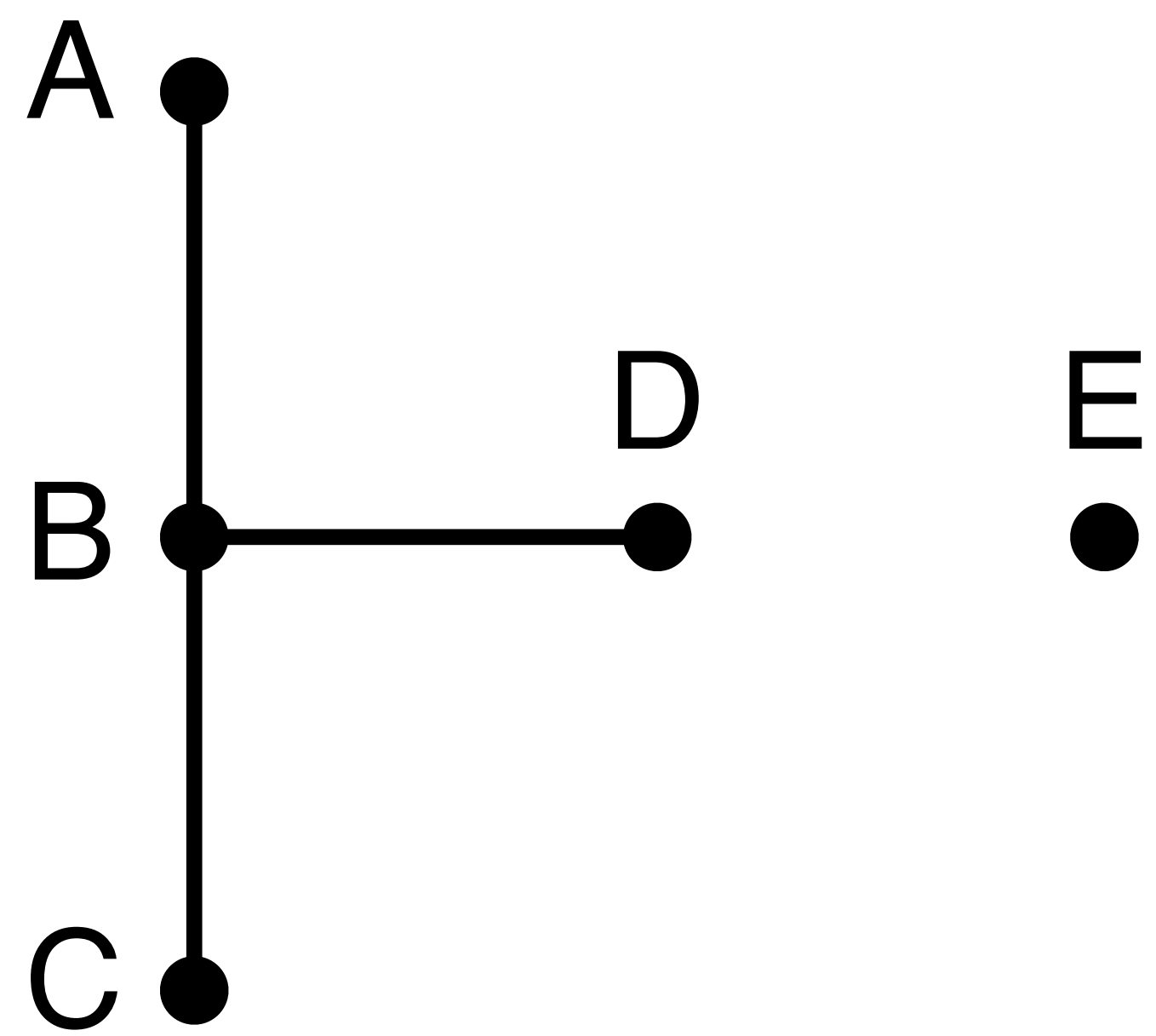
k=2	A	B	C	D	E
A	0	1	0.5	0	0
B	1	0	1	0	0
C	0.5	1	0	1	0.5
D	0	0	1	0	1
E	0	0	0.5	1	0

Interaction-based W: Flows



	A	B	C	D	E
A	0	1	2	1	0
B	1	0	1	0	5
C	2	1	0	1	1
D	1	0	1	0	2
E	0	4	1	3	0

The structure can be a network

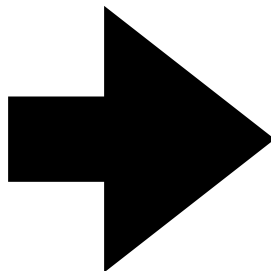


$$w_{ij} = \frac{1}{l_{ij}}$$

	A	B	C	D	E
A	0	1	0.5	0.5	0
B	1	0	1	1	0
C	0.5	1	0	0.5	0
D	0.5	1	0.5	0	0
E	0	0	0	0	0

It is common to standardize W: divide all by sum of row

	A	B	C	D	E
A	0	1	2	1	0
B	1	0	1	0	5
C	2	1	0	1	1
D	1	0	1	0	2
E	0	4	1	3	0



	A	B	C	D	E	Σ
A	0	1/4	1/2	1/4	0	1
B	1/7	0	1/7	0	5/7	1
C	2/5	1/5	0	1/5	1/5	1
D	1/4	0	1/4	0	1/2	1
E	0	1/2	1/8	3/8	0	1

The choice of W should reflect the studied interactions

Spreading processes like COVID

Contiguity, Flow

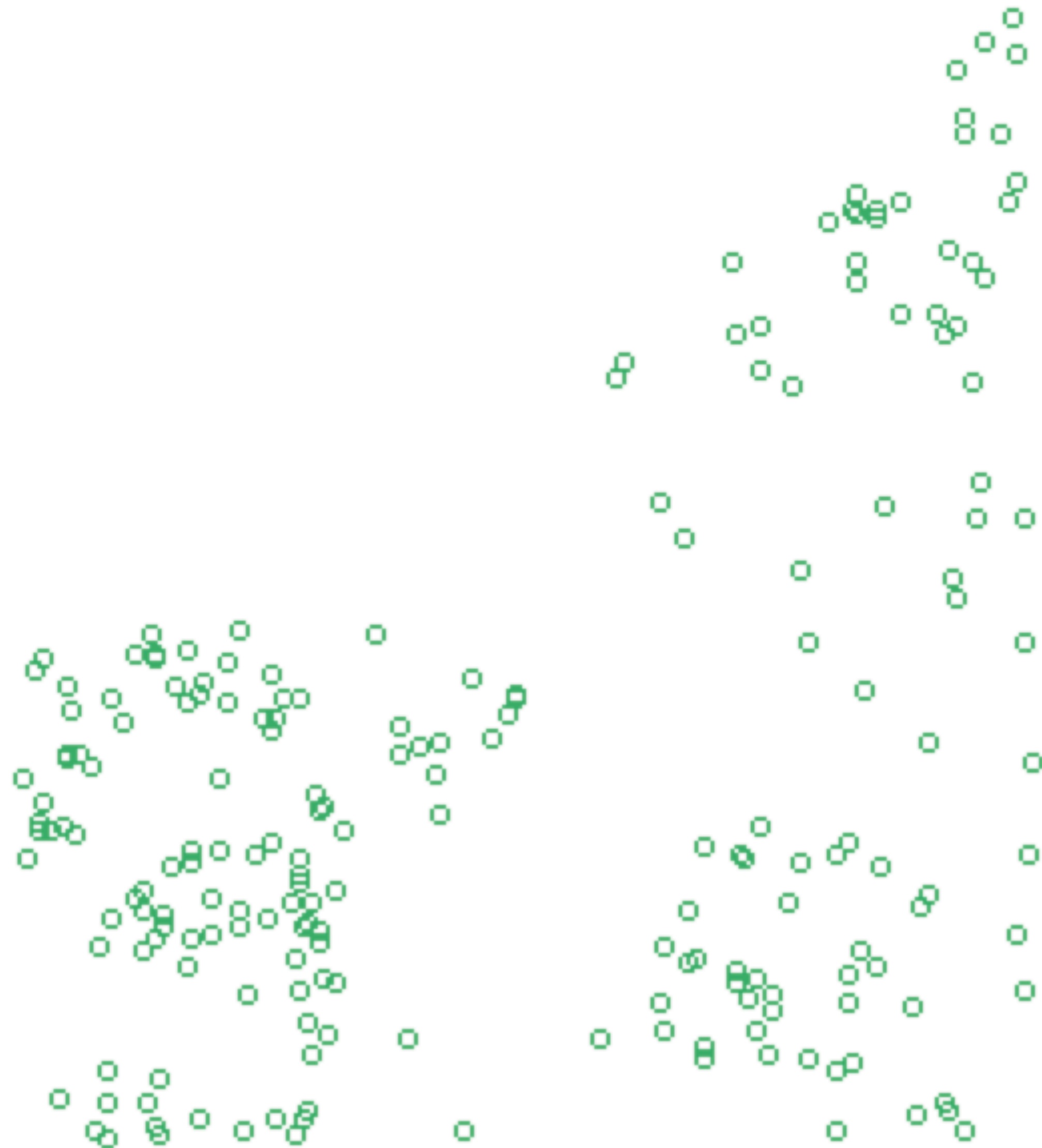
Accessibility

Distance

Effects of laws on counties

Block

Distance-bands are sensitive to cut-off distance



Distance-bands are sensitive to cut-off distance

Connectivity graph with
default cut-off can lead to
"too many" connections

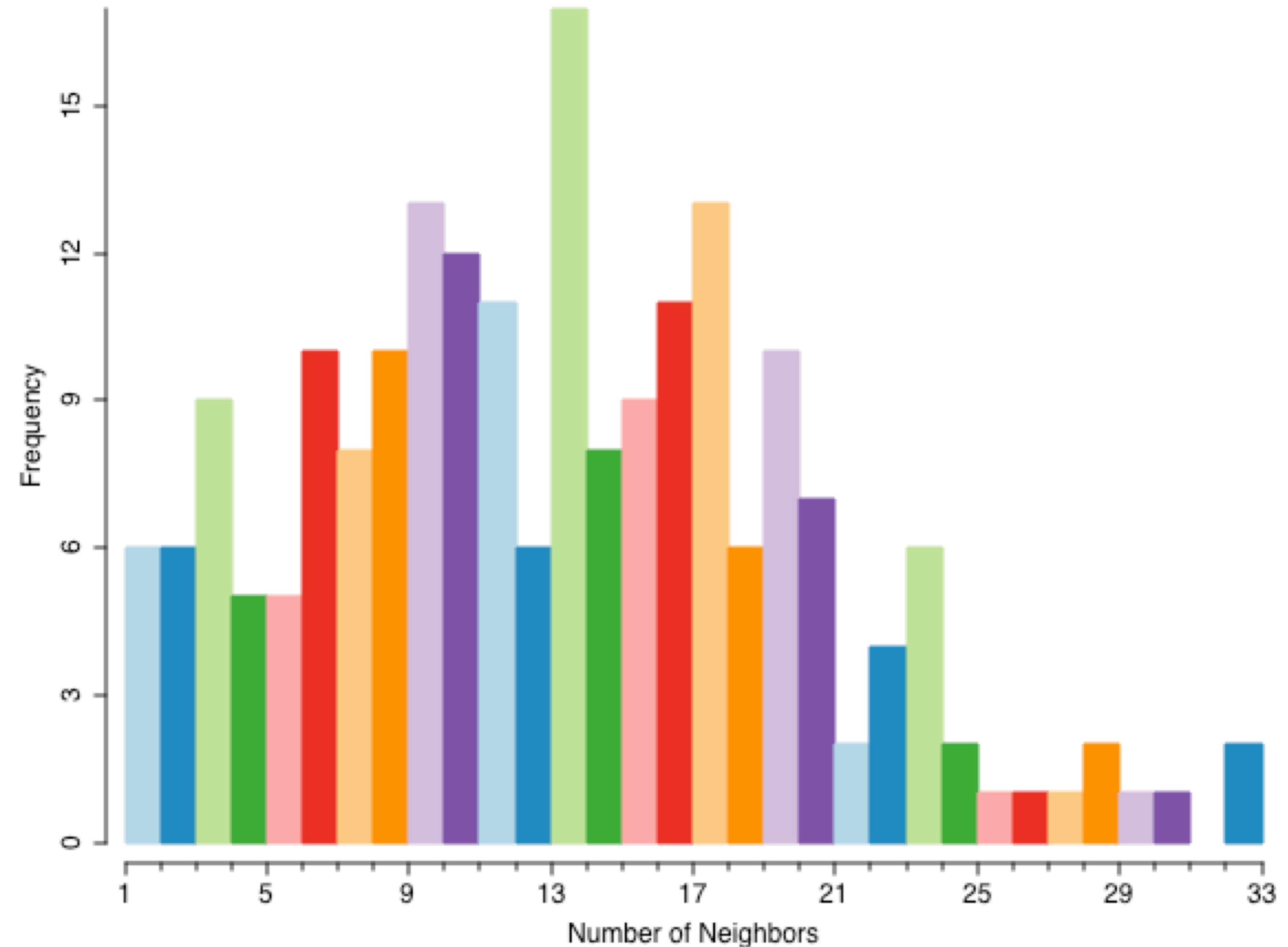


Distance-bands are sensitive to cut-off distance

Connectivity graph with default cut-off can lead to "too many" connections

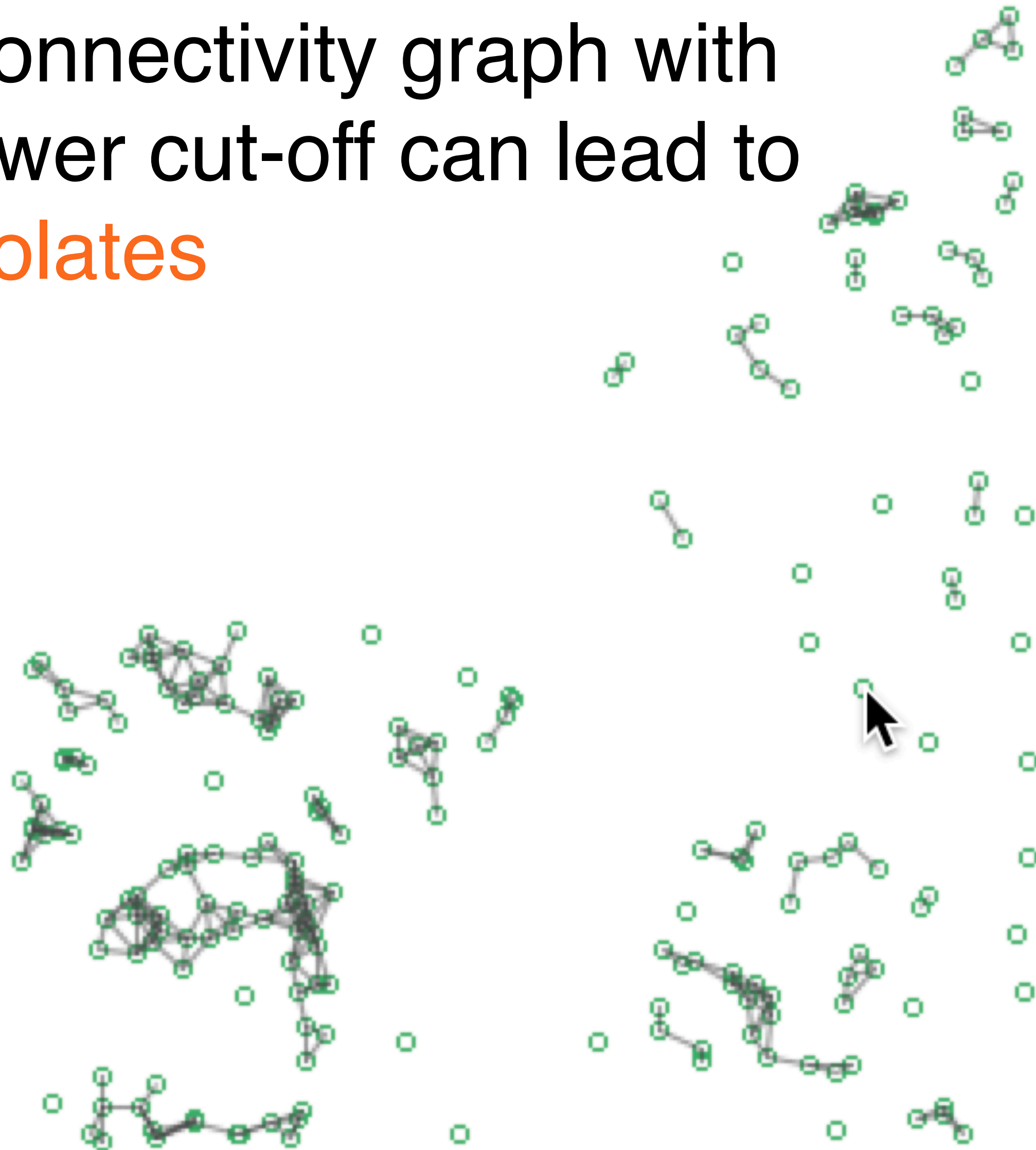


Degree distribution is called:
Connectivity histogram

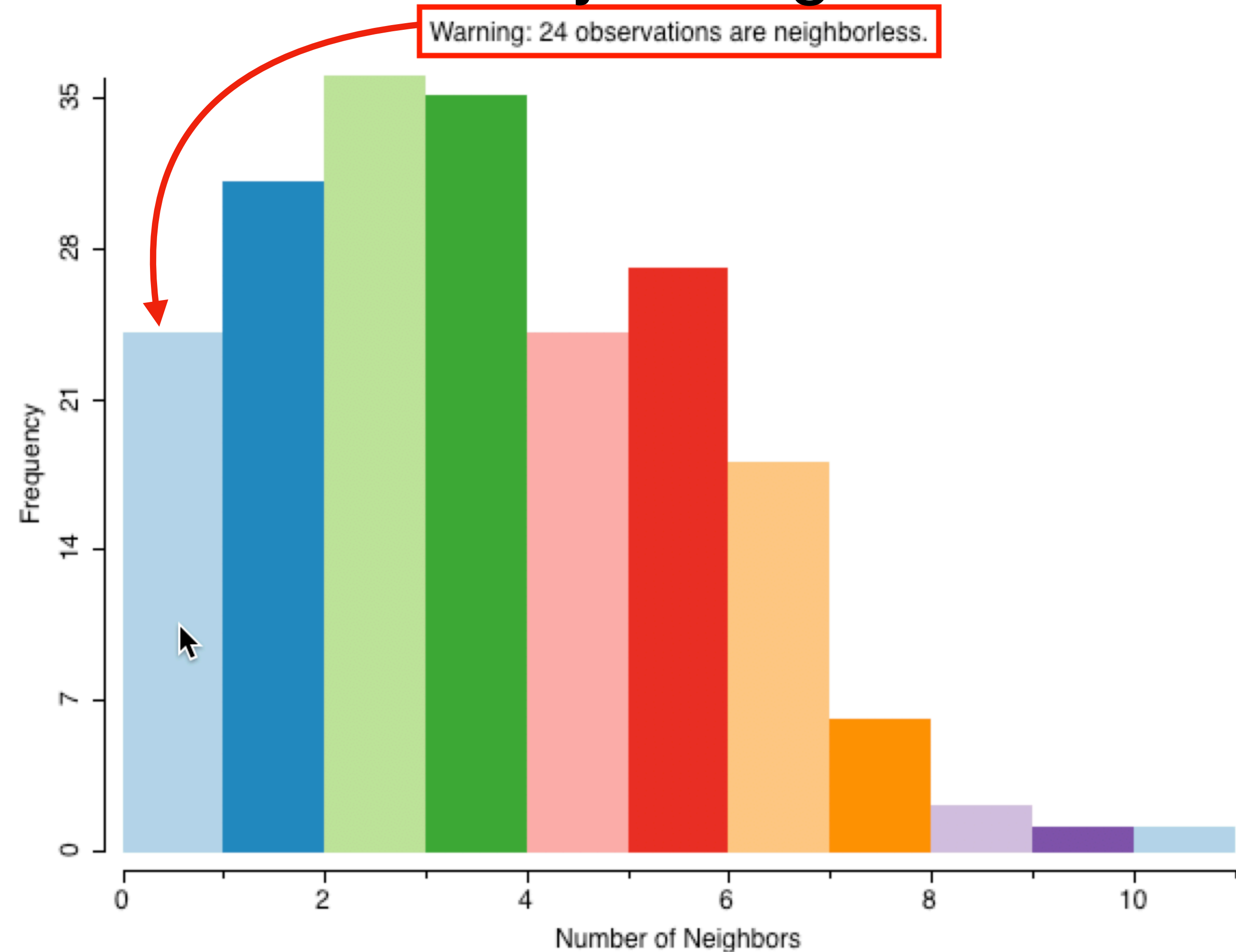


Distance-bands are sensitive to cut-off distance

Connectivity graph with lower cut-off can lead to **isolates**

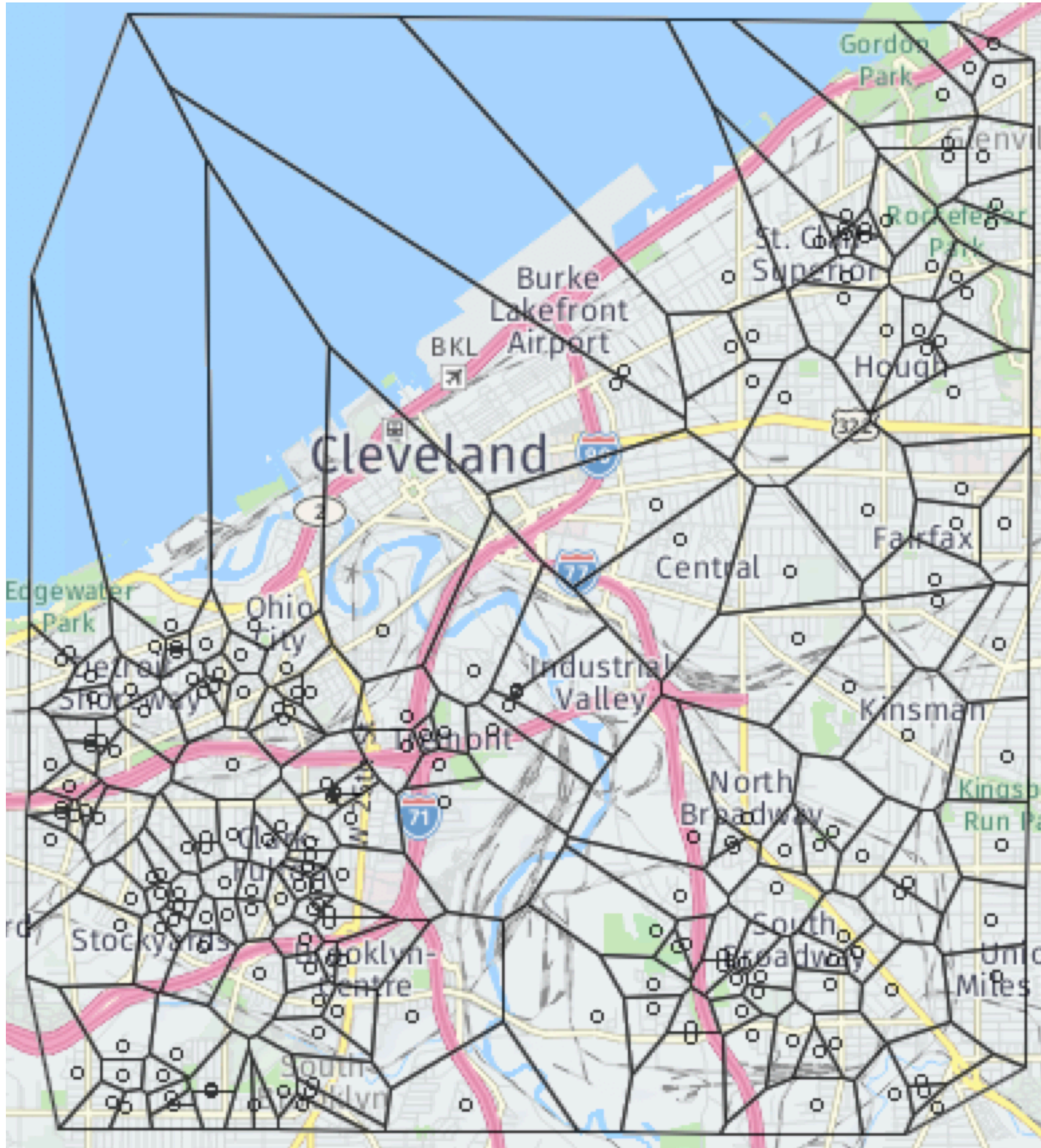


Connectivity histogram

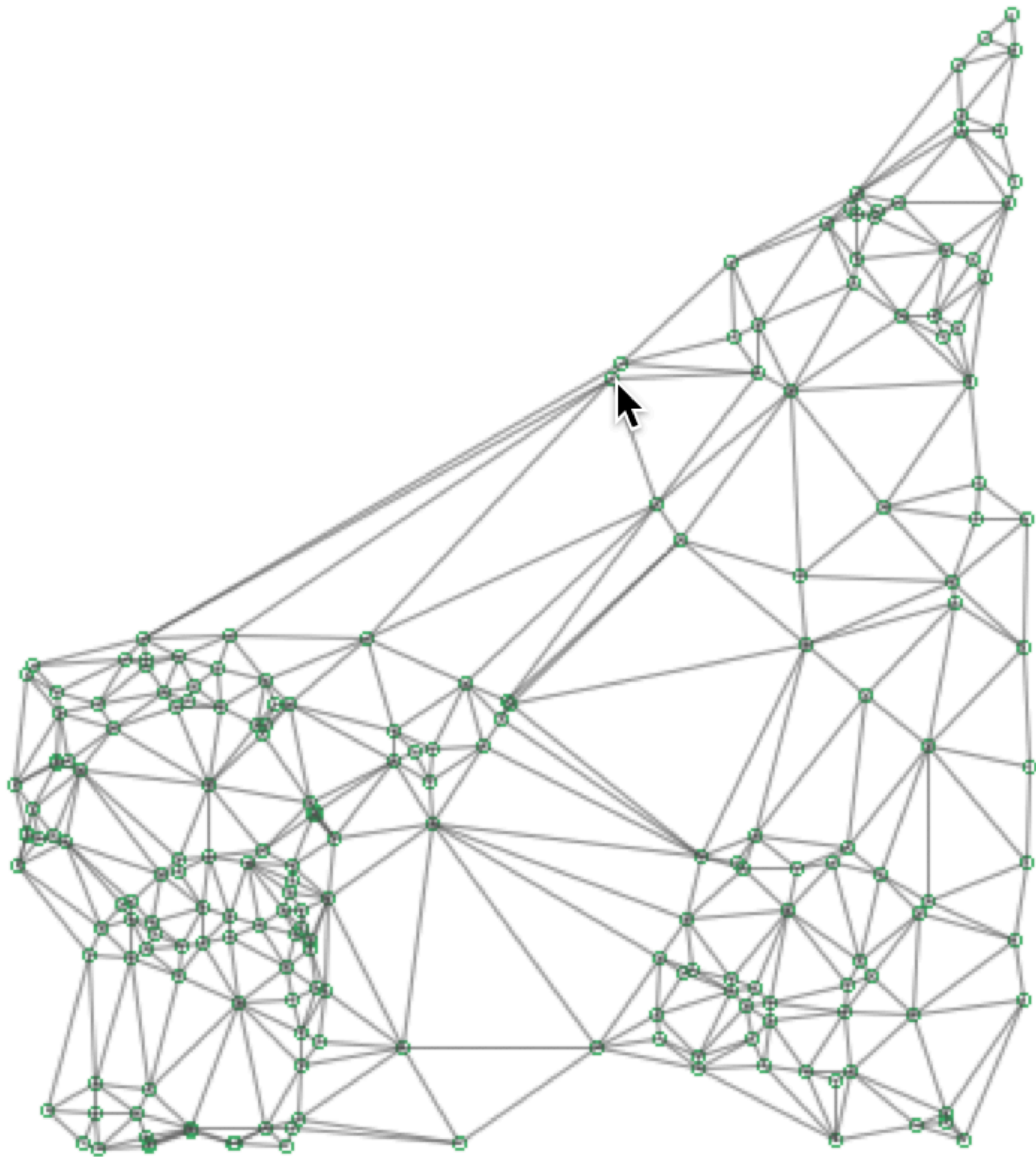


Thiessen polygons* can generalize contiguity

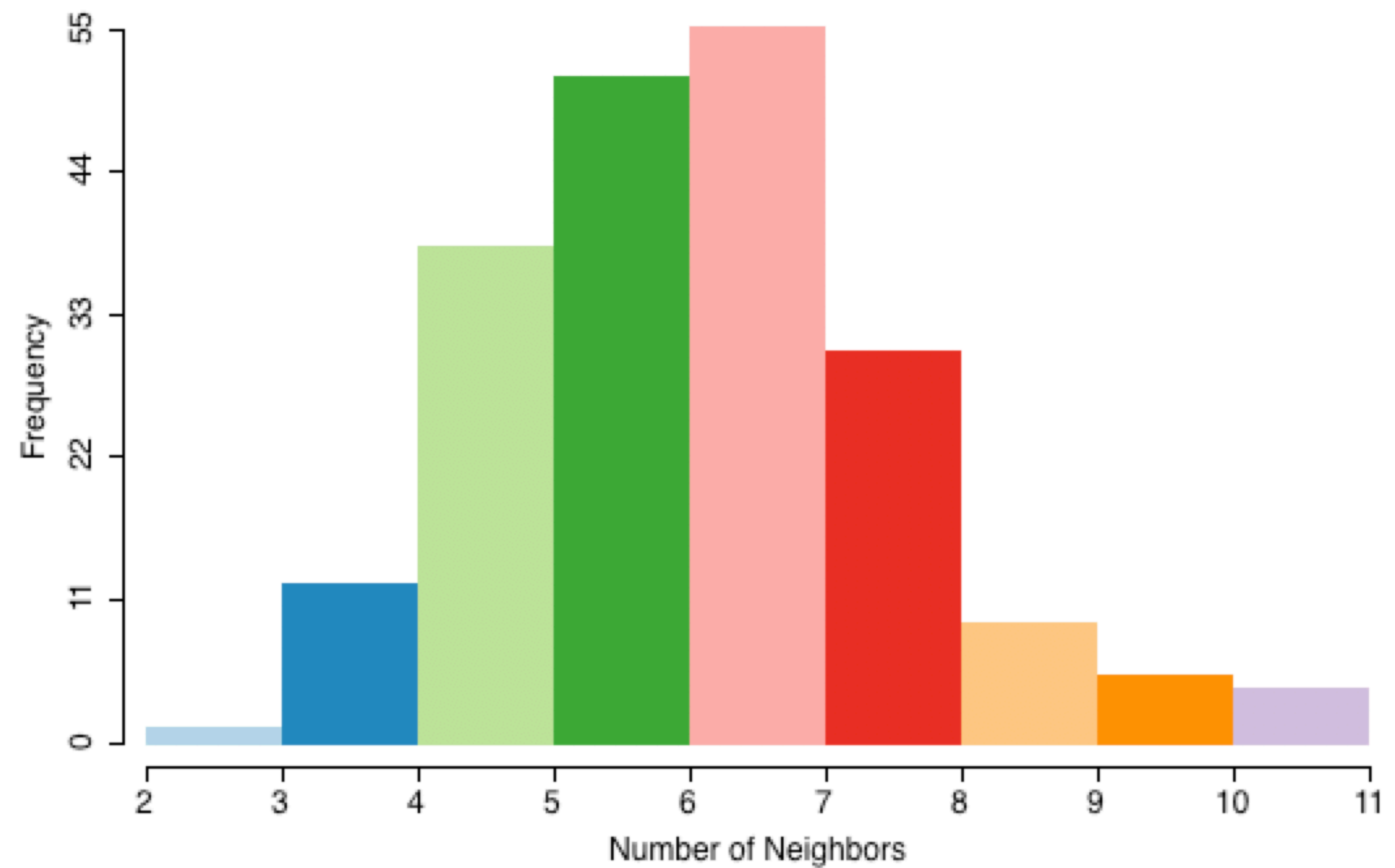
*also called Voronoi cells



Thiessen polygons can generalize contiguity

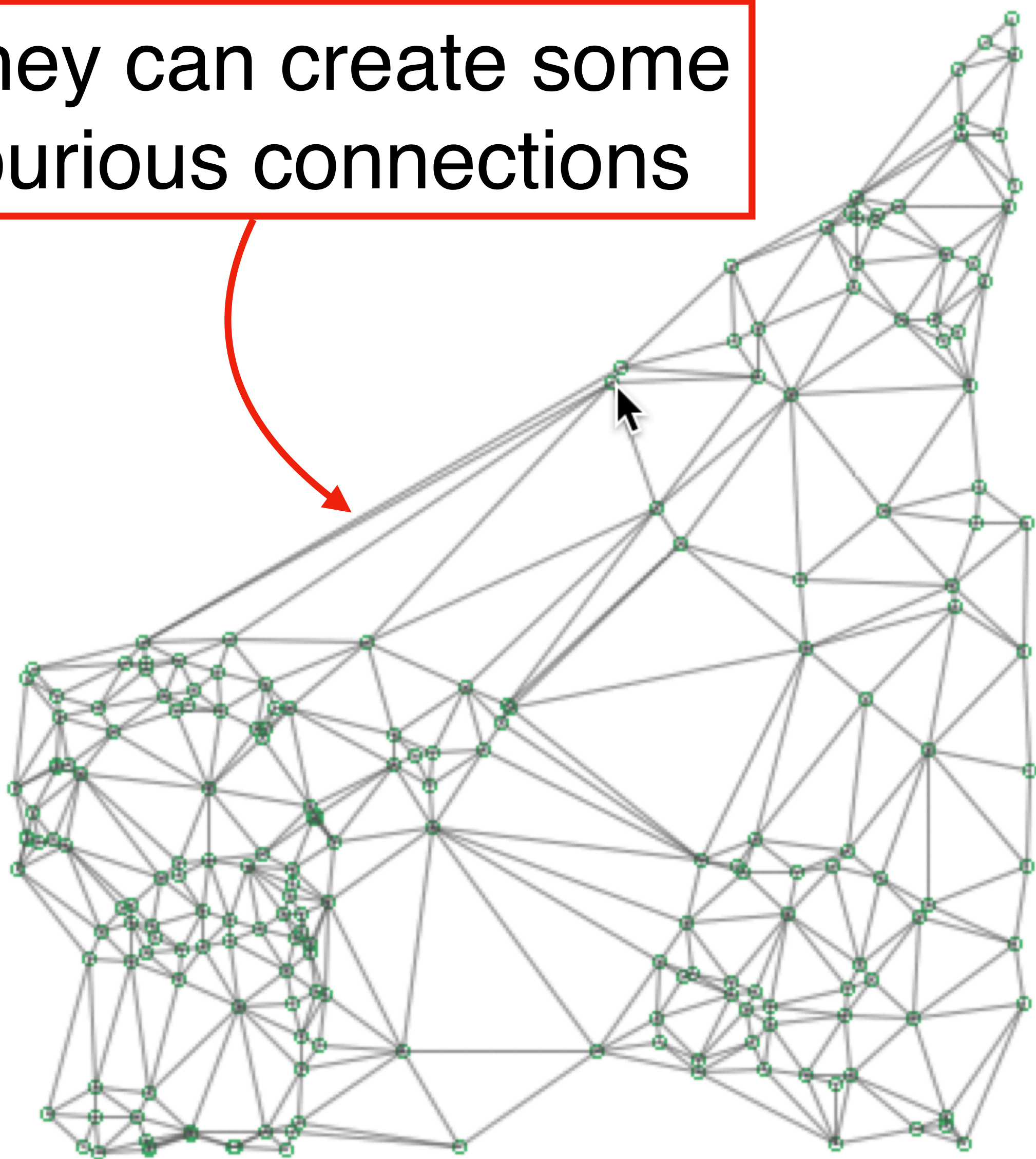


Connectivity histogram

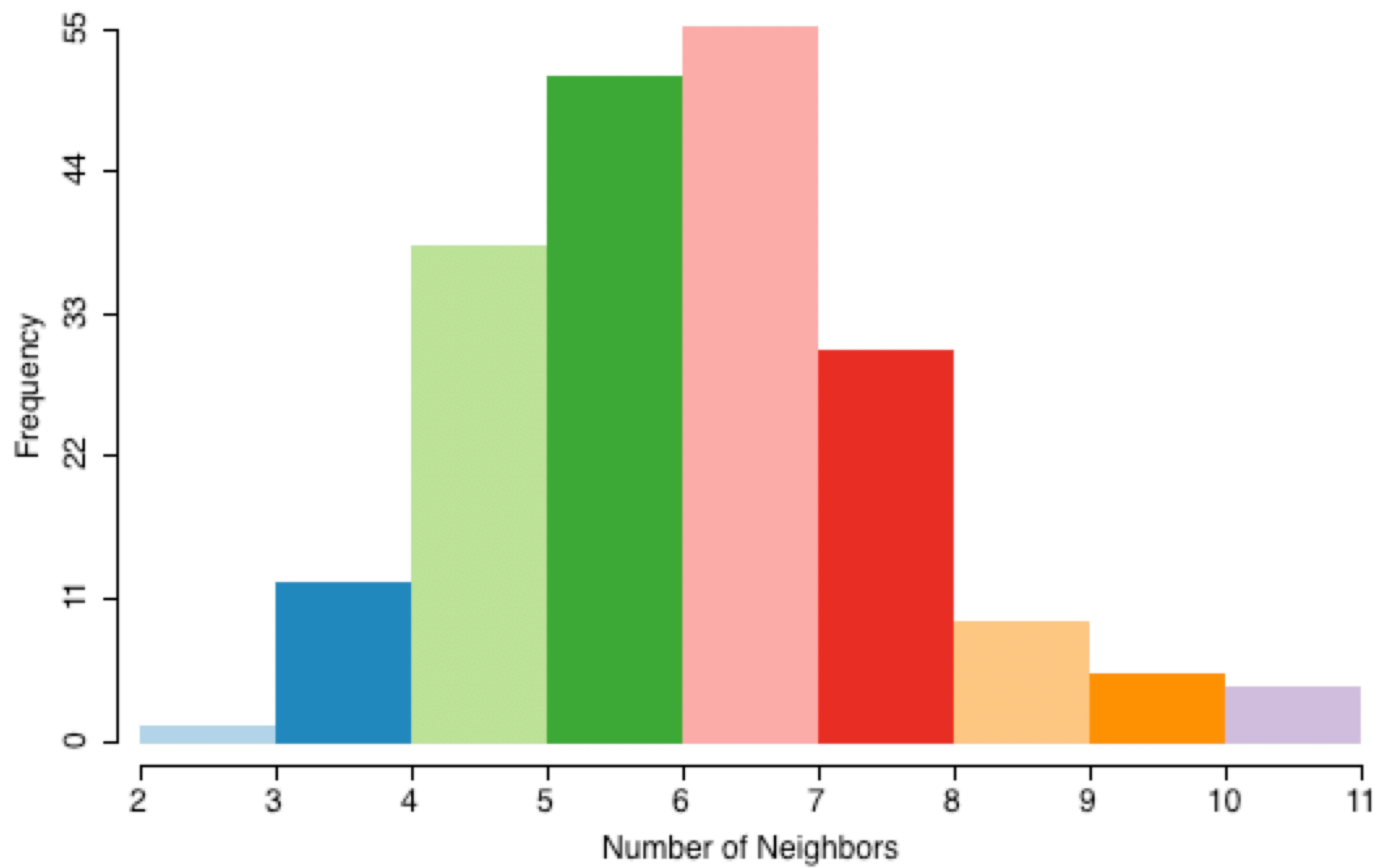


Thiessen polygons can generalize contiguity

They can create some spurious connections

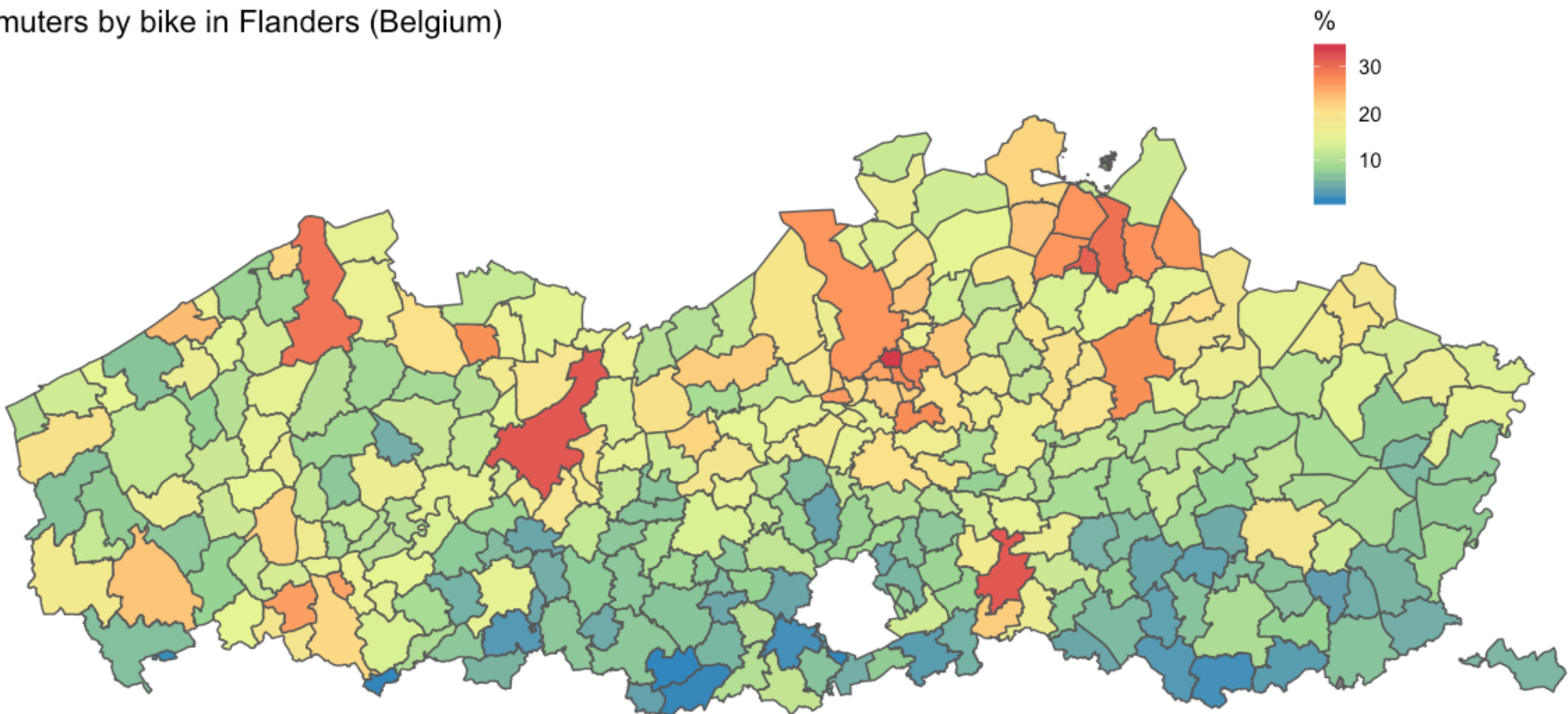


Connectivity histogram

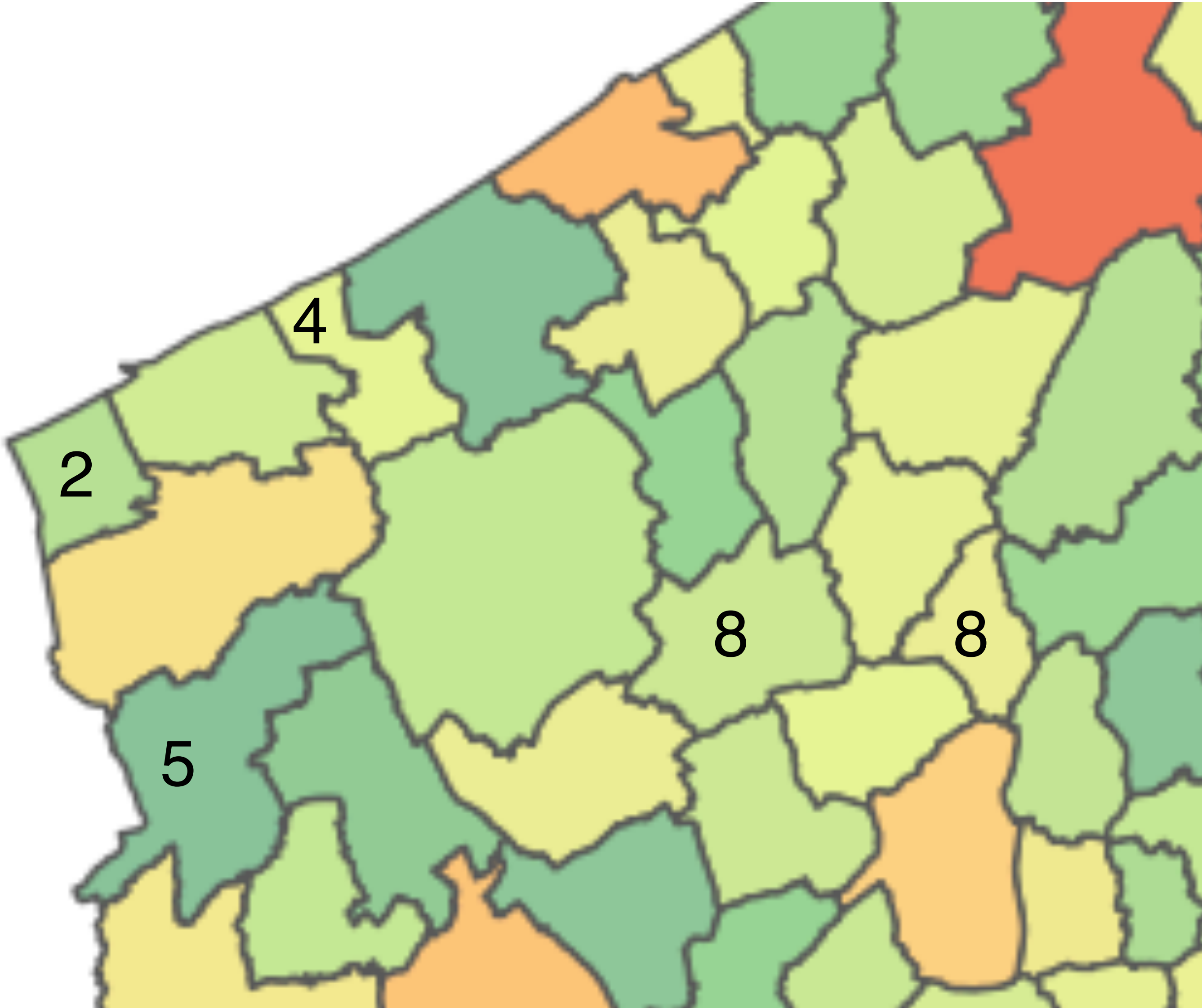


It is also important to consider edge effects!

Commuters by bike in Flanders (Belgium)

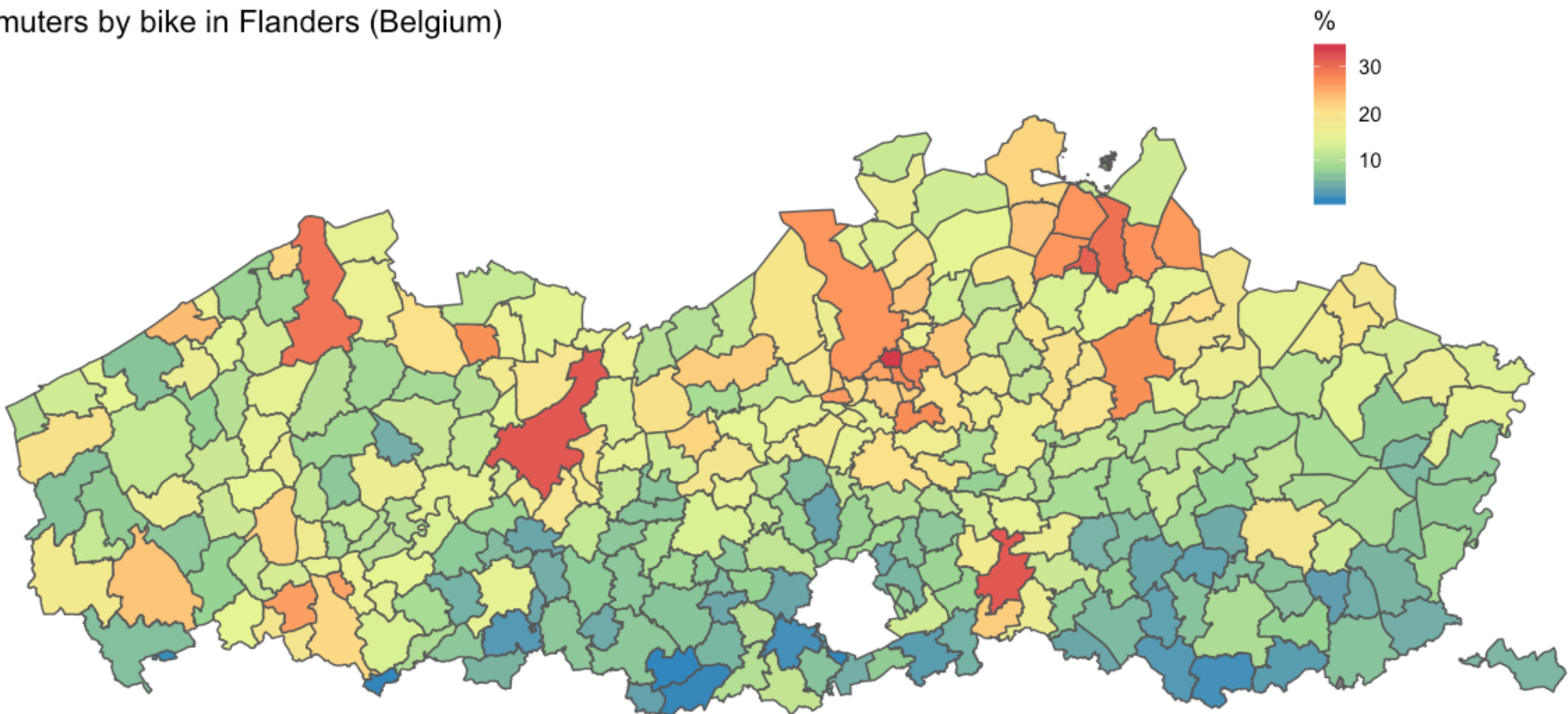


Data: Gemeentemonitor 2017

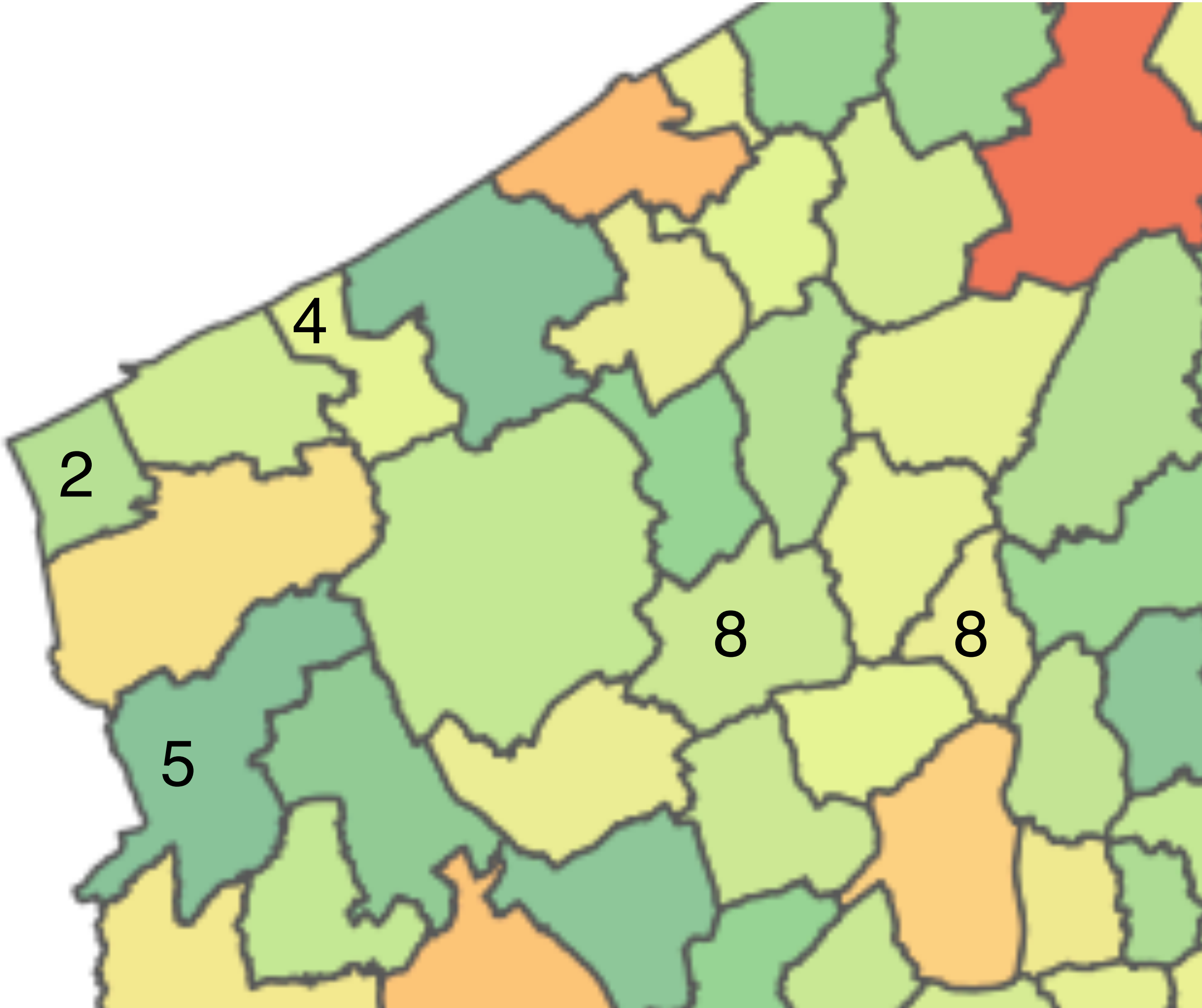
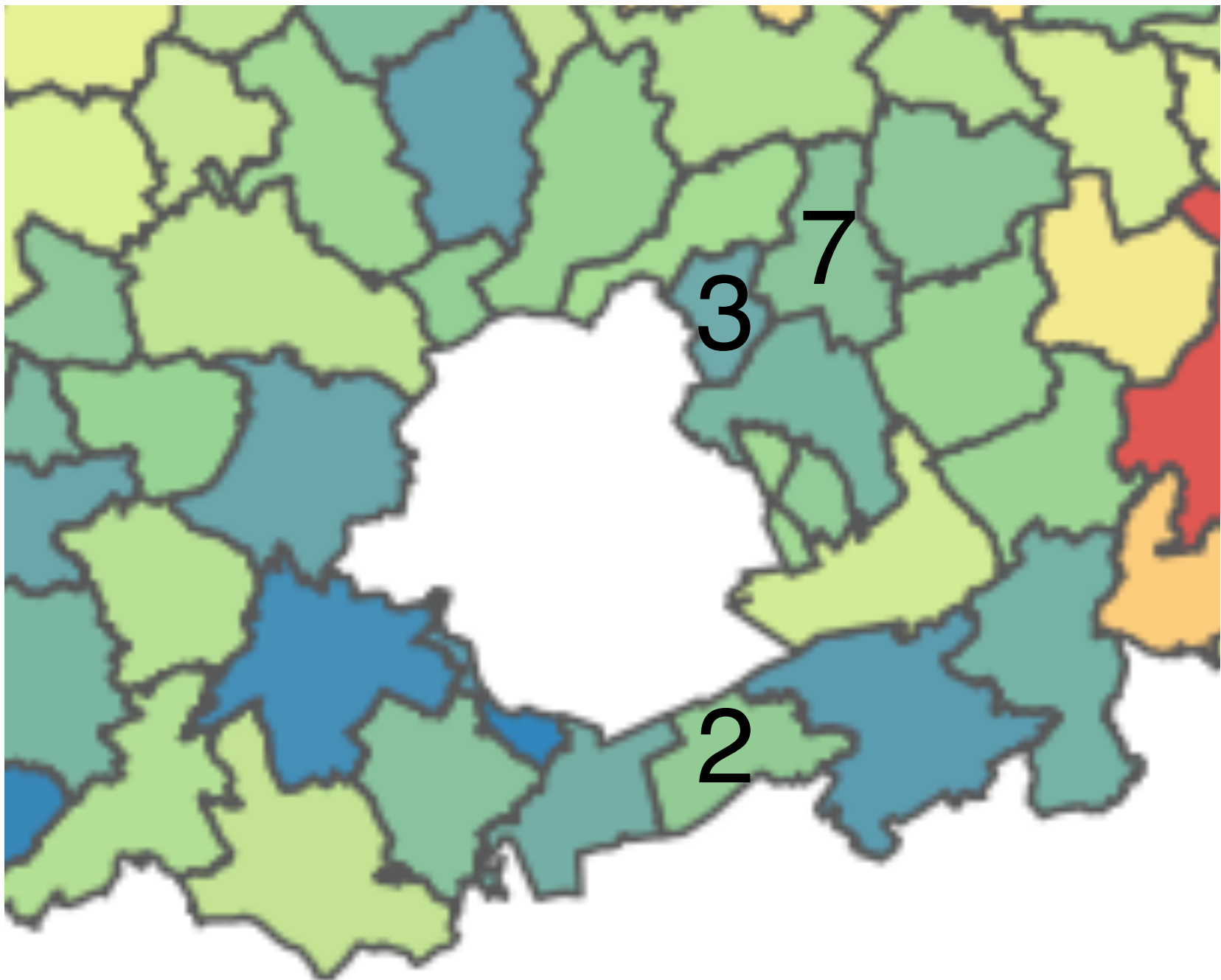


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Commuters by bike in Flanders (Belgium)



Data: Gemeentemonitor 2017

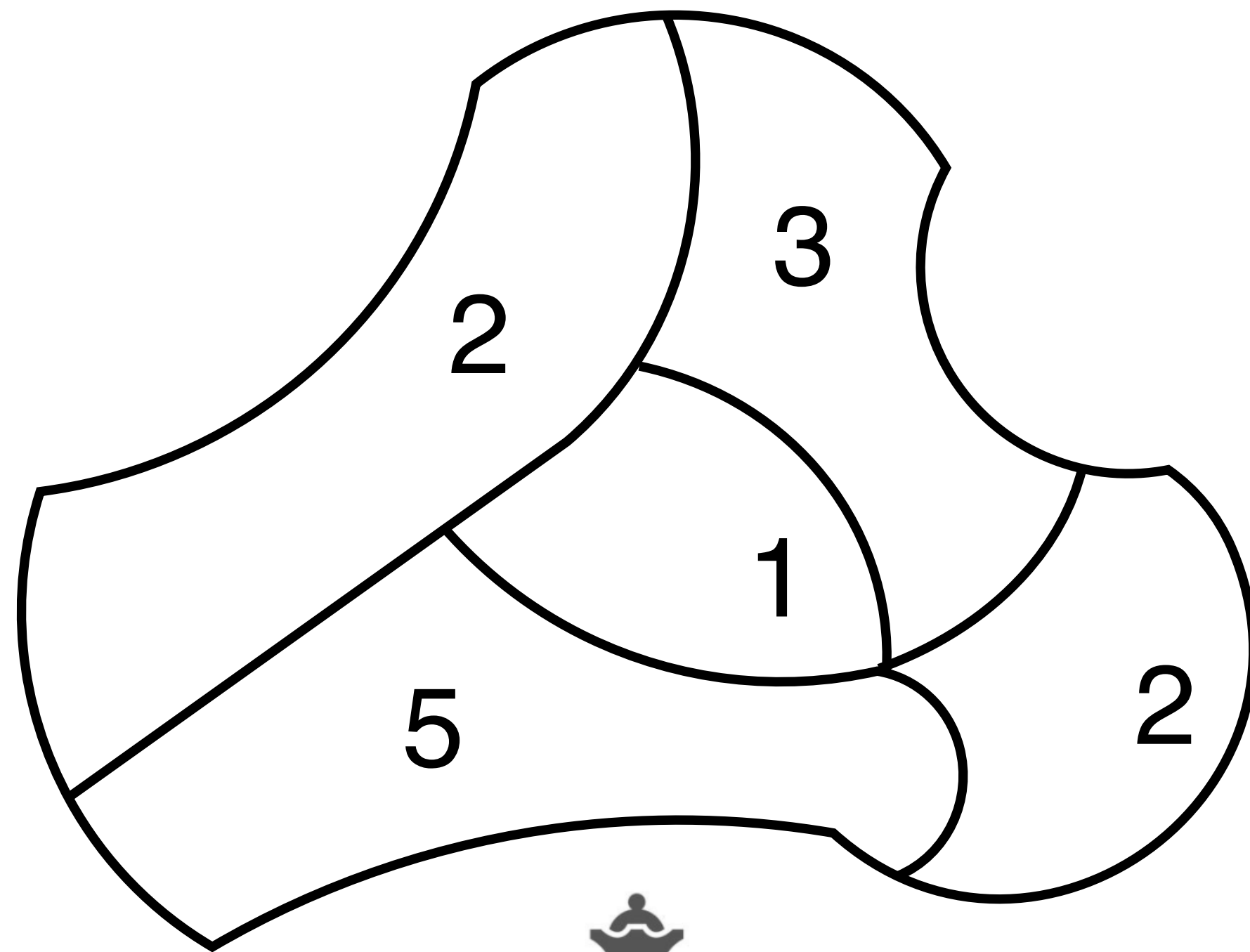


Spatial lag

The spatial lag is the weighted average value of neighbors

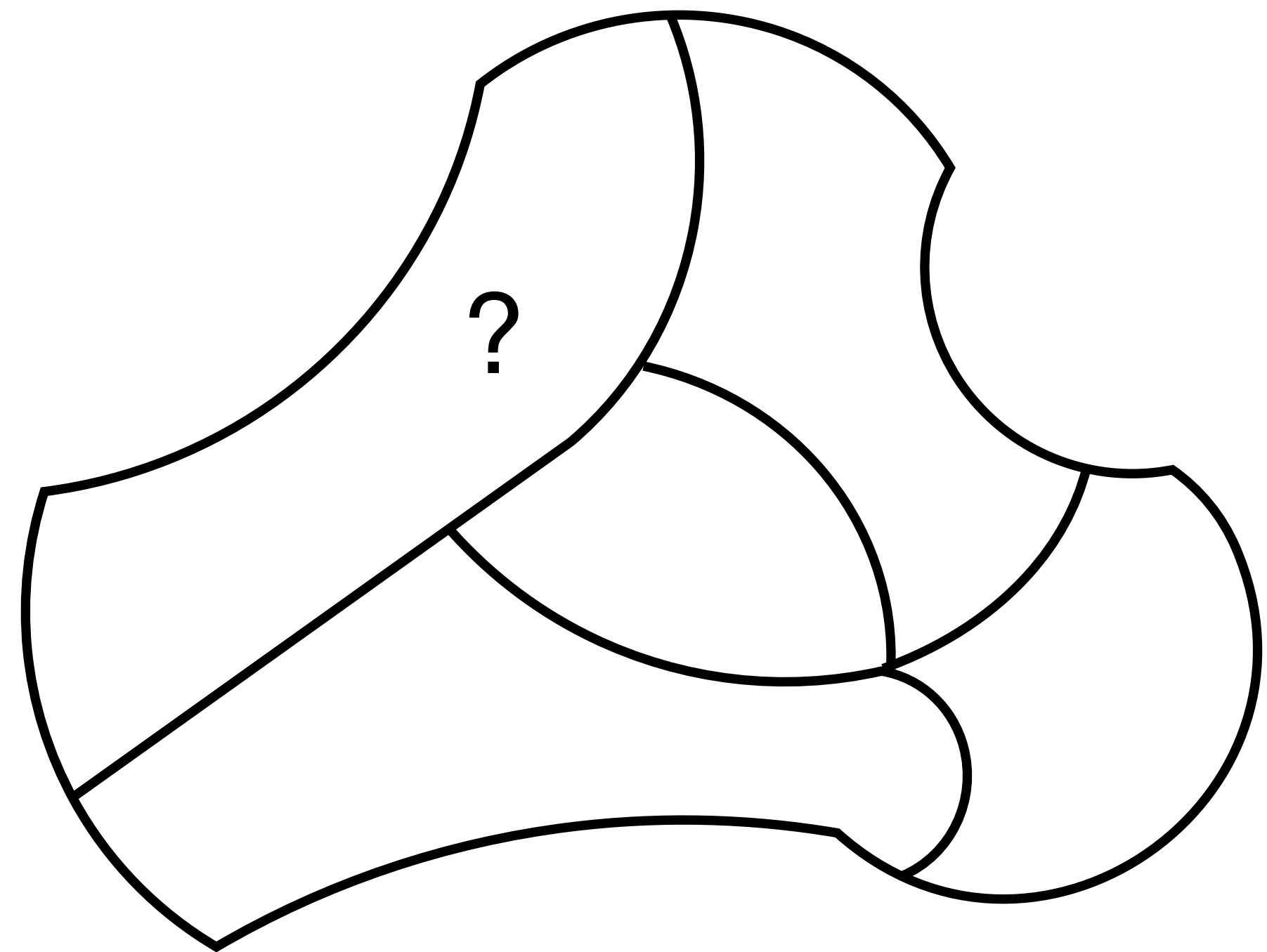
if W is standardized

Values y



QUEEN

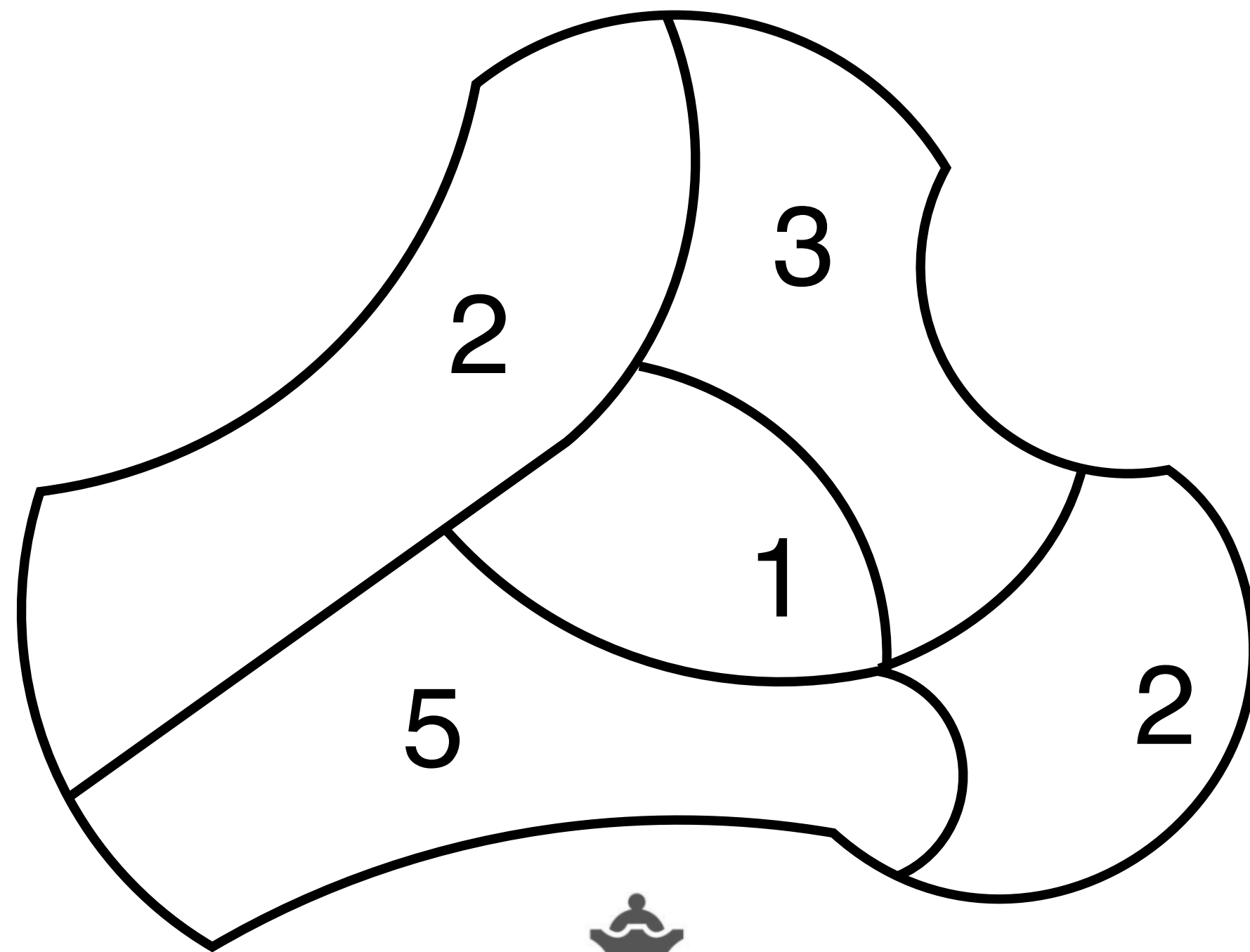
Spatial lag y_{lag}



The spatial lag is the weighted average value of neighbors

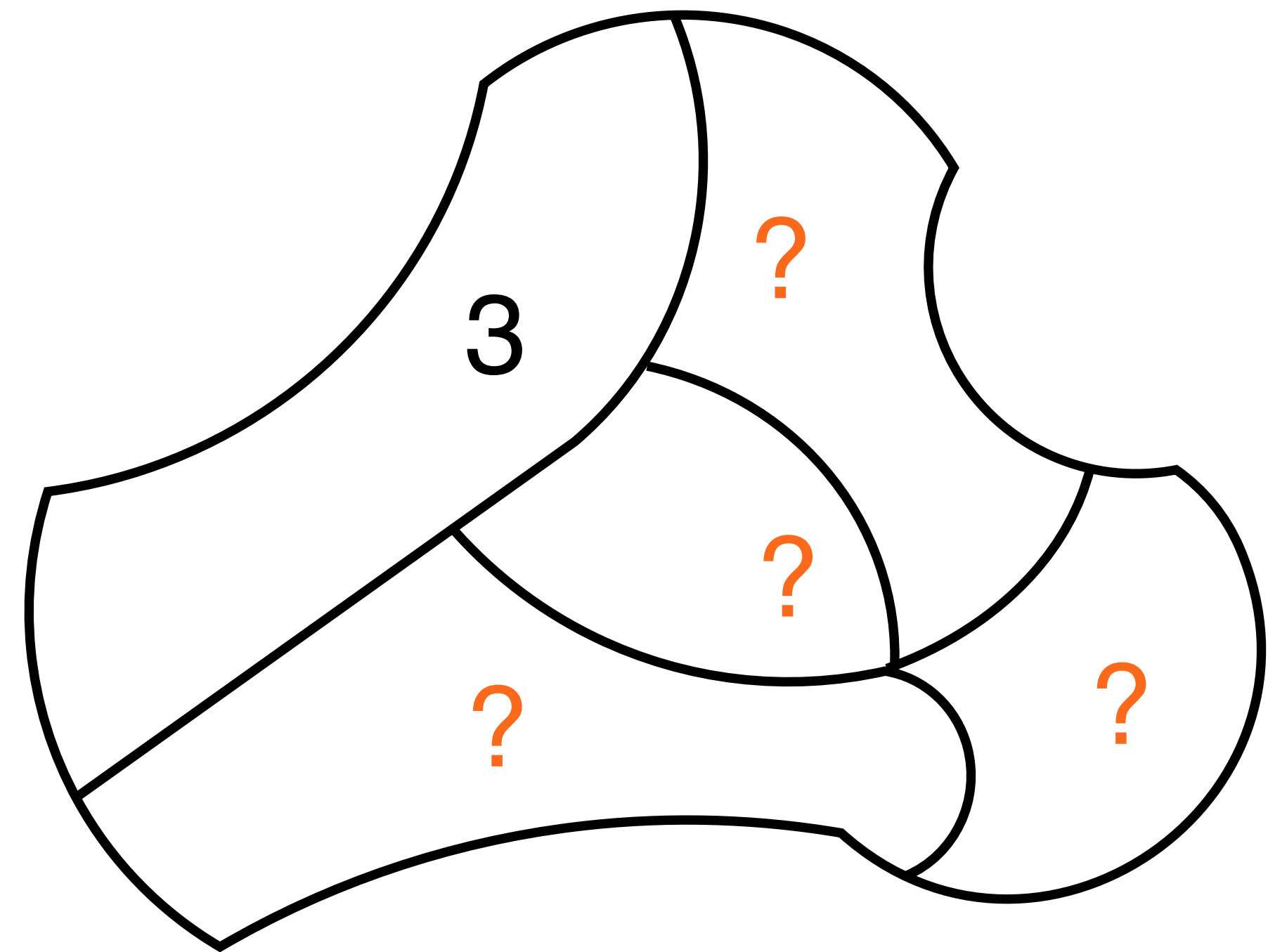
if W is standardized

Values y



QUEEN

Spatial lag y_{lag}

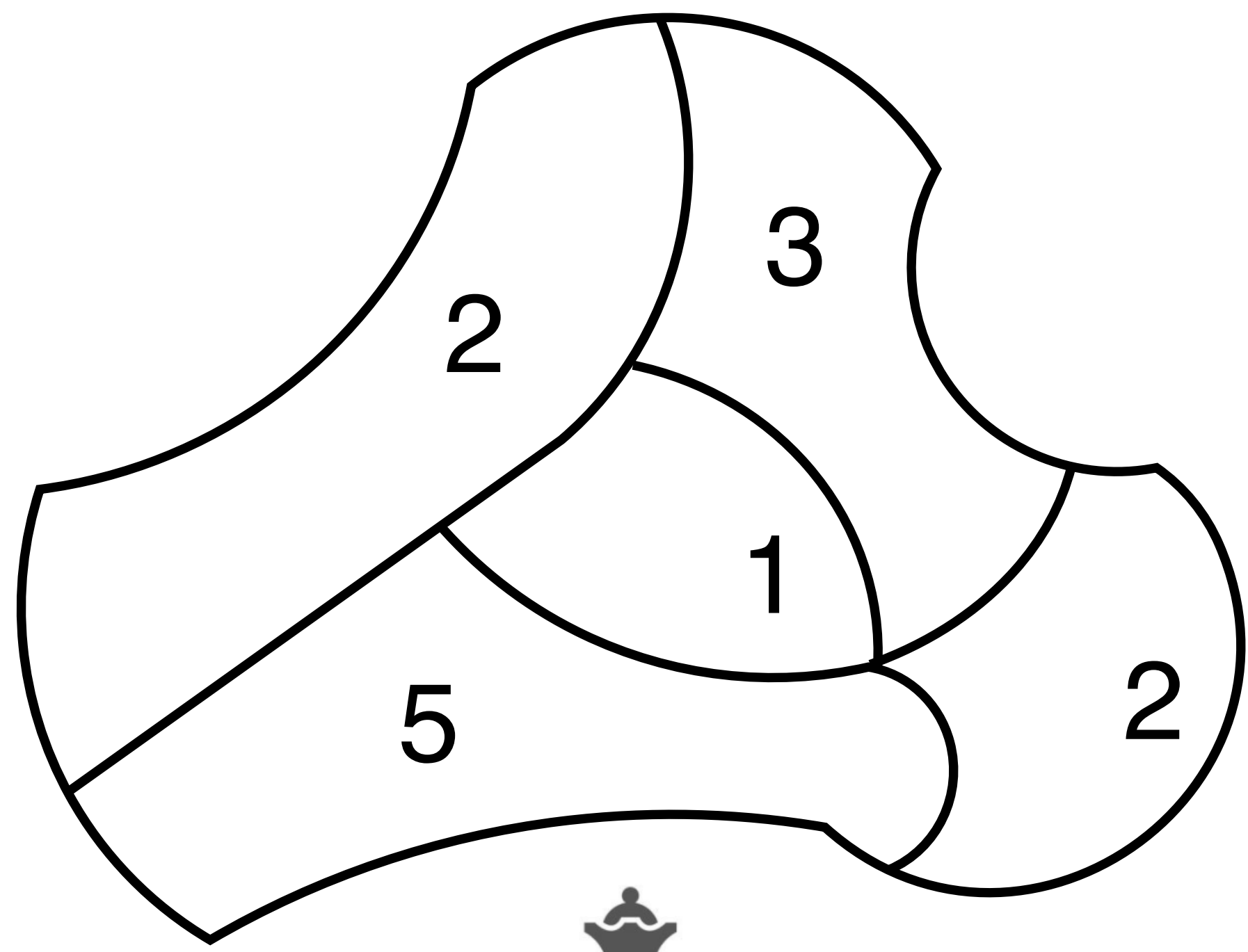


Exercise

The spatial lag is the weighted average value of neighbors

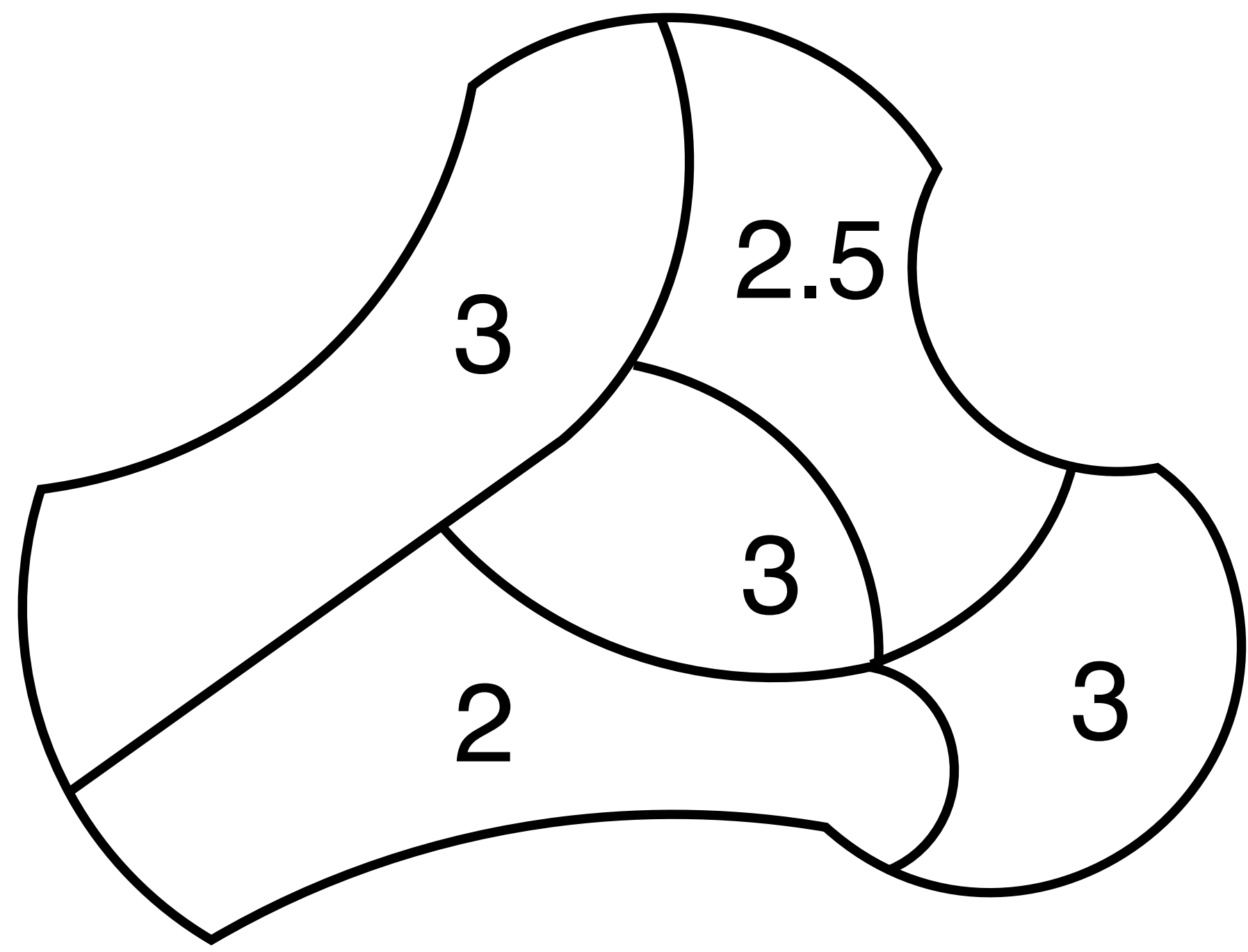
if W is standardized

Values y



QUEEN

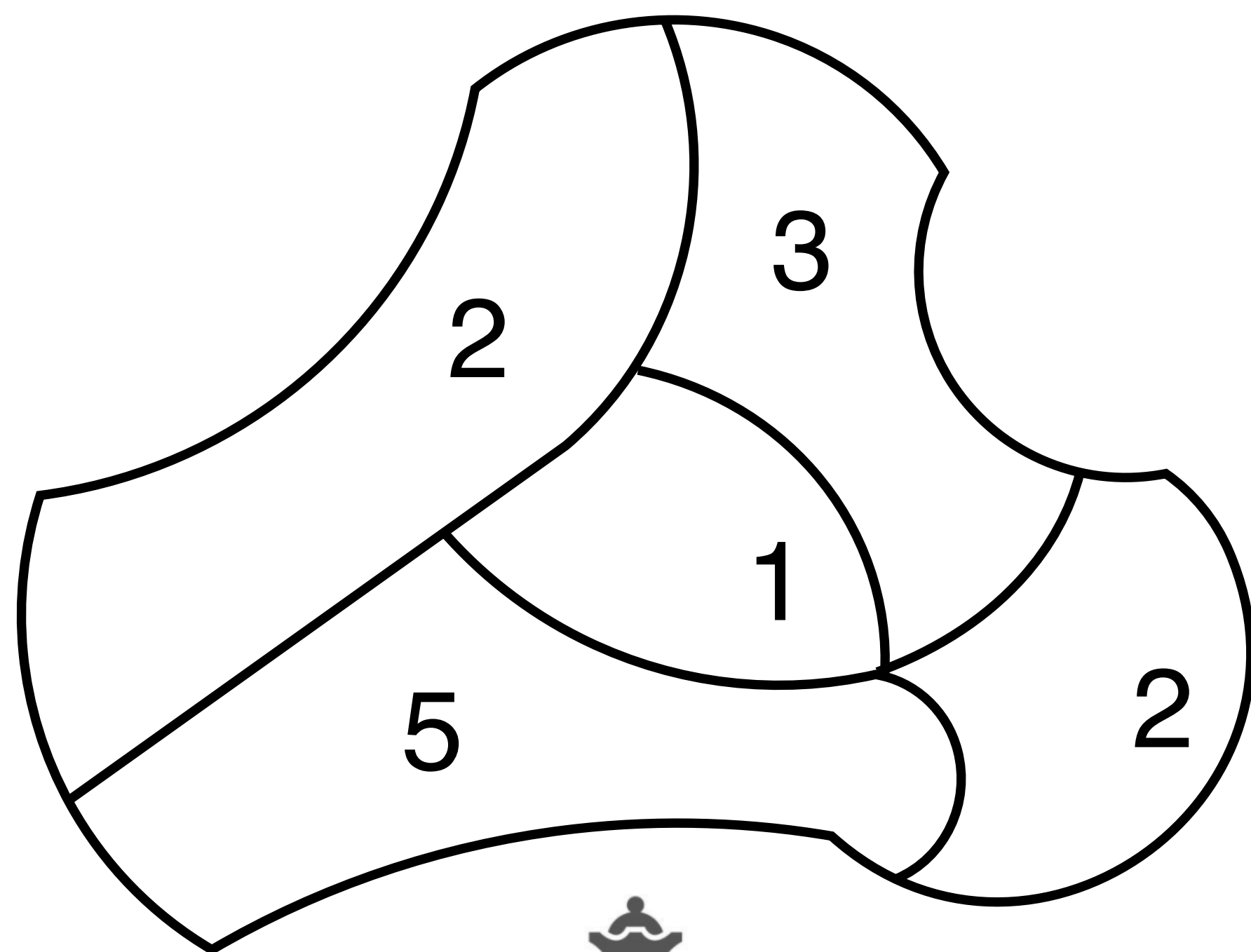
Spatial lag y_{lag}



The spatial lag is the weighted average value of neighbors

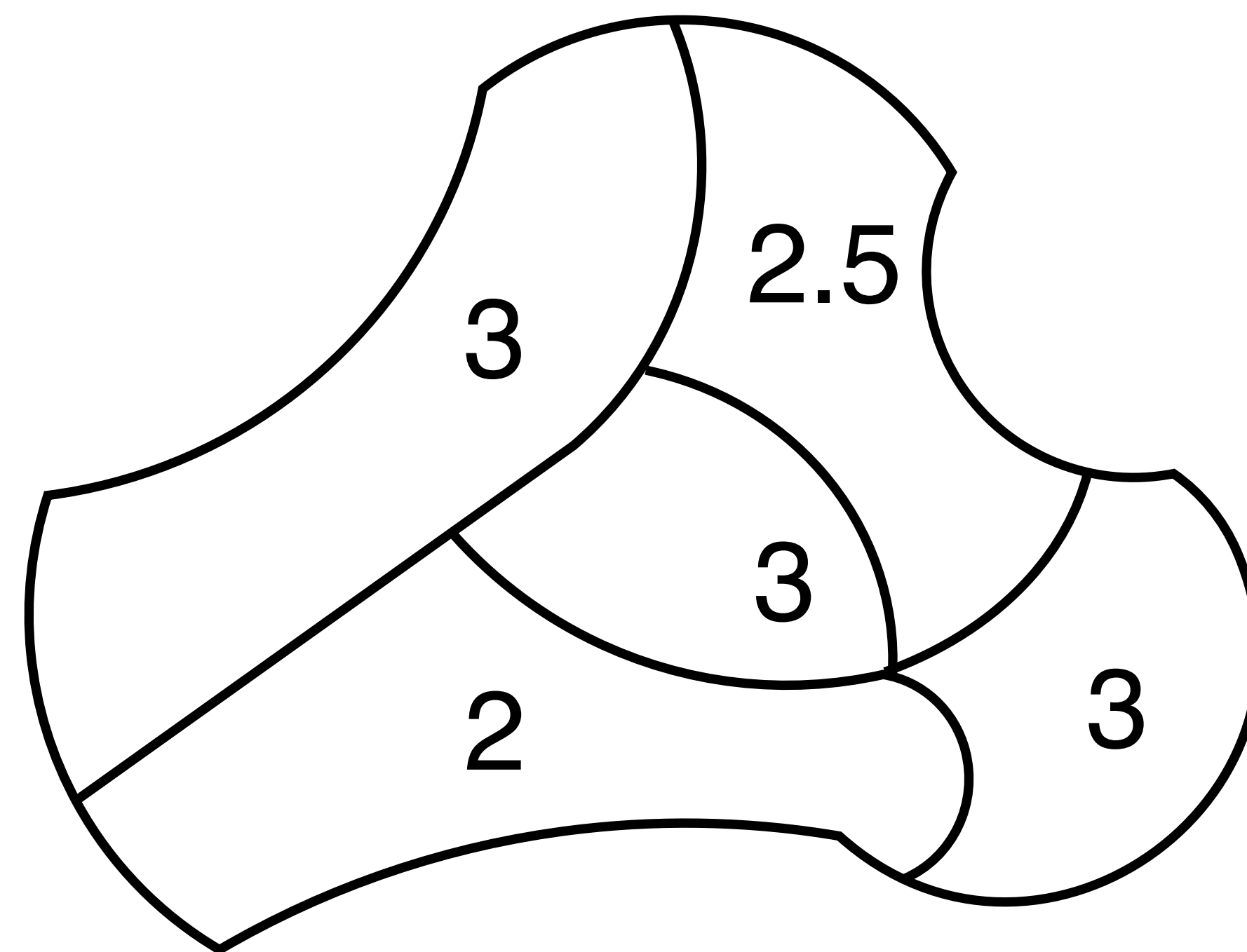
if W is standardized

Values y



QUEEN

Spatial lag y_{lag}



It is a smoother: It brings all values closer to the average

The spatial lag is the sum of products of weights and values

$$y_{\text{lag},i} = w_{i1}y_1 + w_{i2}y_2 + \cdots + w_{in}y_n = \sum_{j=1}^n w_{ij}y_j$$

The spatial lag is the sum of products of weights and values

$$y_{\text{lag},i} = w_{i1}y_1 + w_{i2}y_2 + \cdots + w_{in}y_n = \sum_{j=1}^n w_{ij}y_j$$

$$\mathbf{y}_{\text{lag}} = \left(\sum_{j=1}^n w_{ij}y_j \right)_i = W\mathbf{y}$$

The spatial lag is the sum of products of weights and values

$$y_{\text{lag},i} = w_{i1}y_1 + w_{i2}y_2 + \cdots + w_{in}y_n = \sum_{j=1}^n w_{ij}y_j$$

$$\mathbf{y}_{\text{lag}} = \left(\sum_{j=1}^n w_{ij}y_j \right)_i = W\mathbf{y}$$

The spatial lag appears in
many tools and models

We use PySAL to define spatial weights

```
from pysal.lib import weights
```

```
w_queen = weights.Queen.from_dataframe(db, ids="area_id")
```

spatial weight criteria

your GeoDataframe

column to index weights on



We use PySAL to define spatial weights

```
w_queen[ '708518' ]
```

```
{ '706412': 1.0,  
  '700870': 1.0,  
  '709938': 1.0,  
  '707896': 1.0,  
  '718916': 1.0,  
  '703634': 1.0,  
  '711110': 1.0,  
  '701970': 1.0,  
  '703720': 1.0 }
```

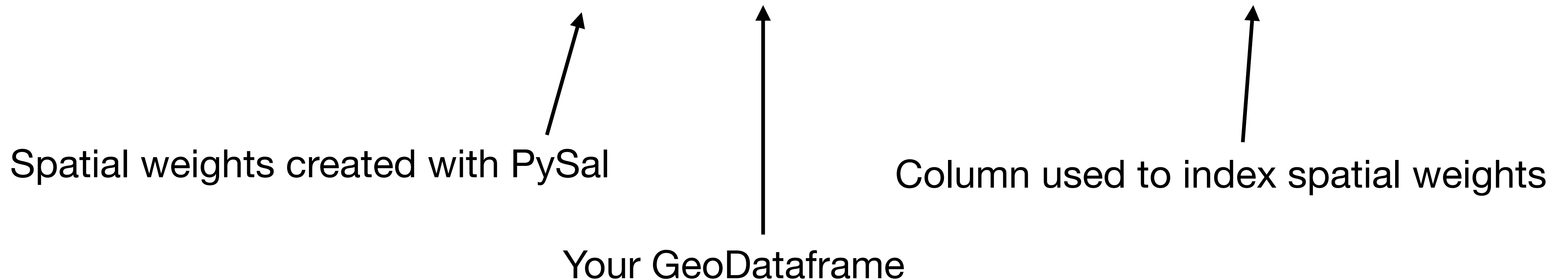


New library: splot

```
from splot.libpysal import plot_spatial_weights
```

```
plot_spatial_weights(w_queen, db, indexed_on="area_id")
```

Spatial weights created with PySal



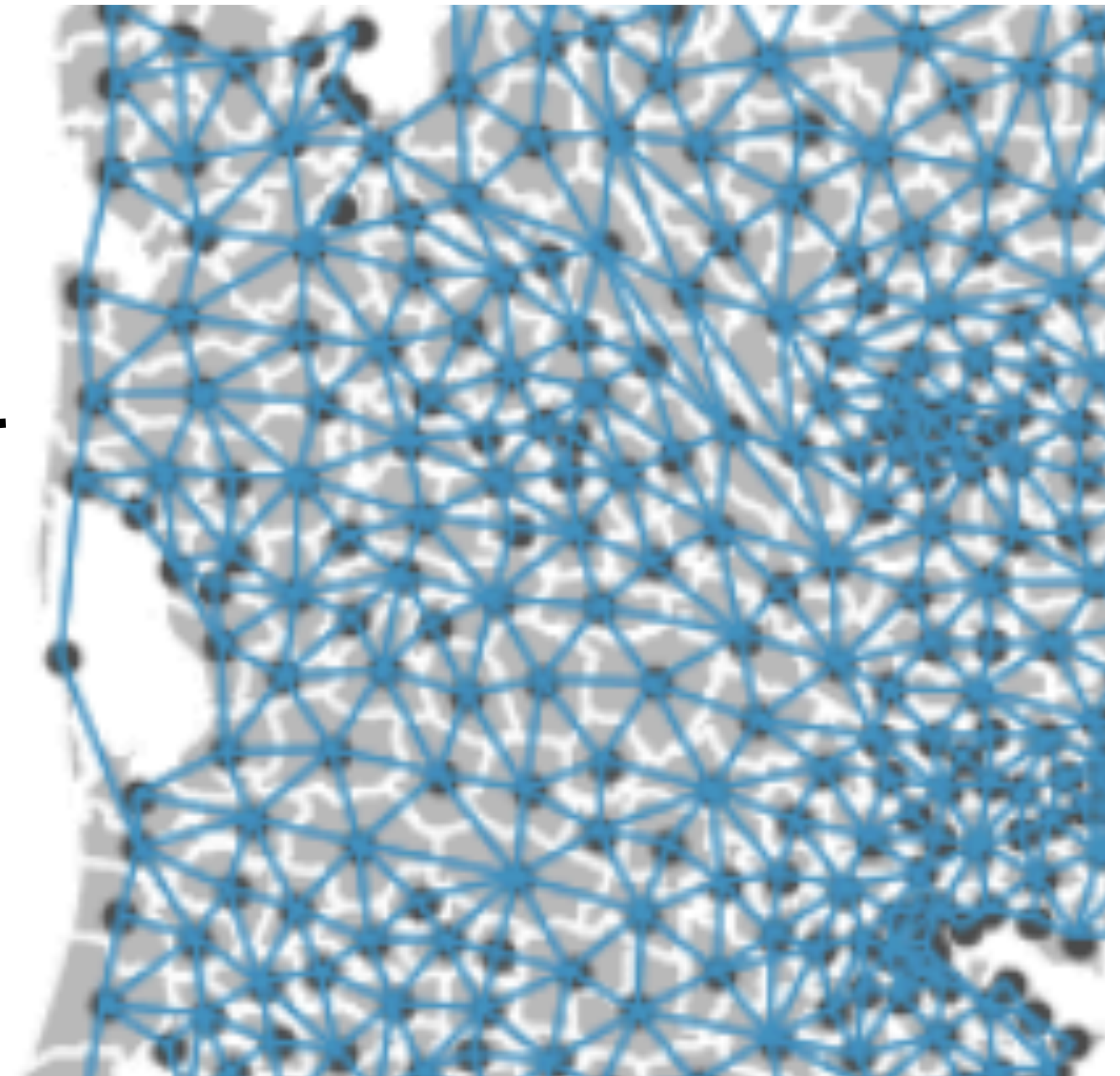
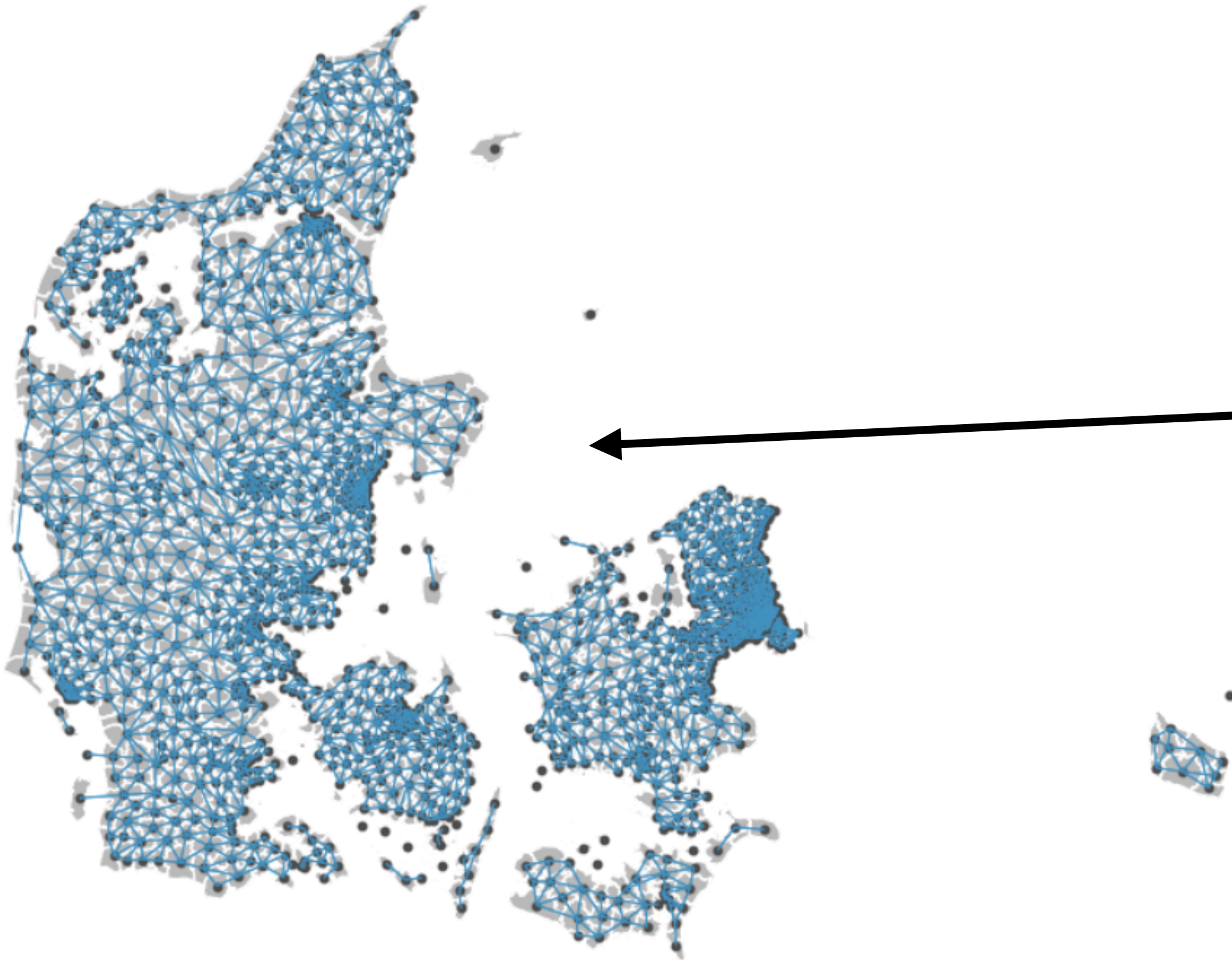
Your GeoDataframe

Column used to index spatial weights

New library: splot

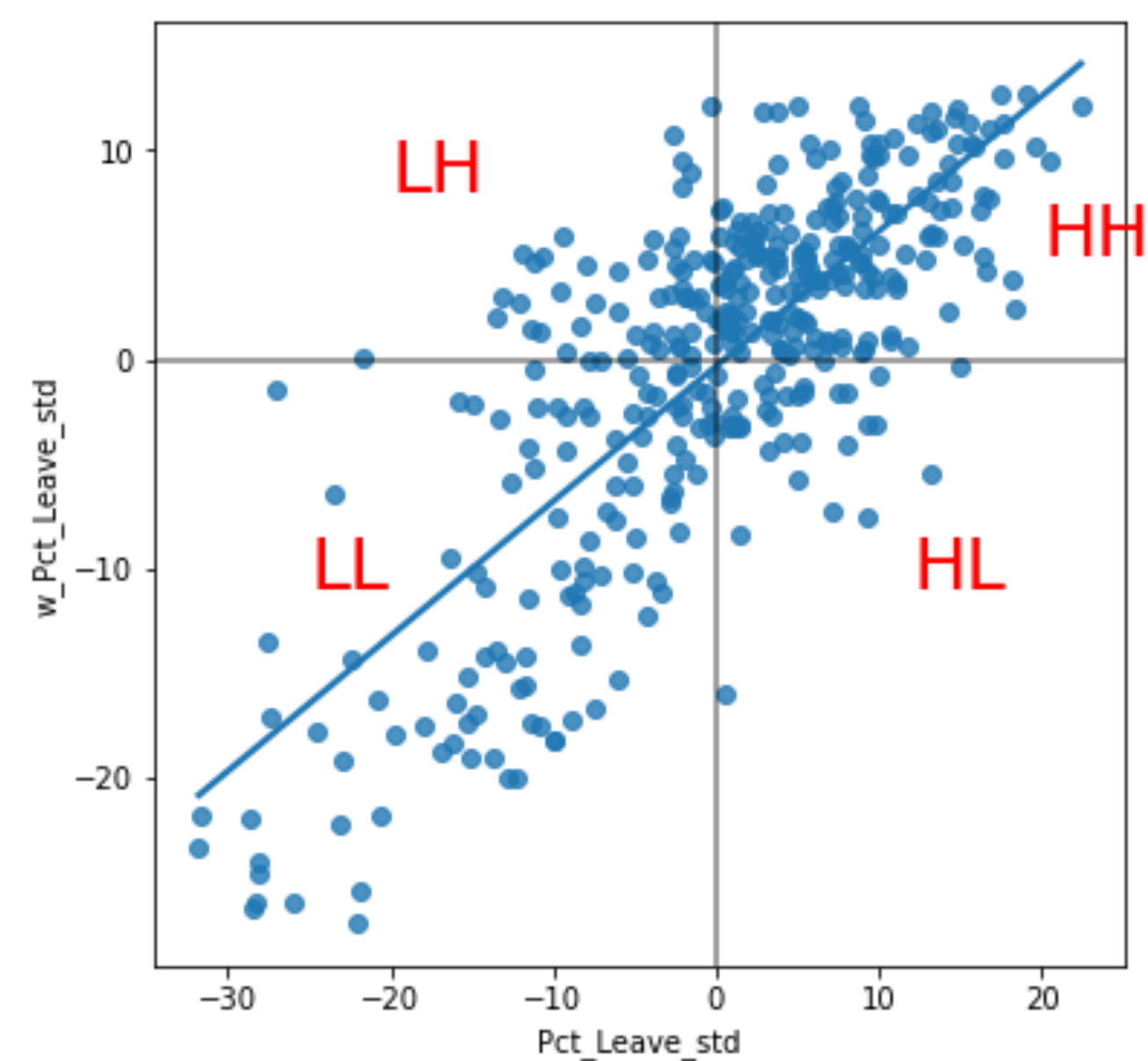
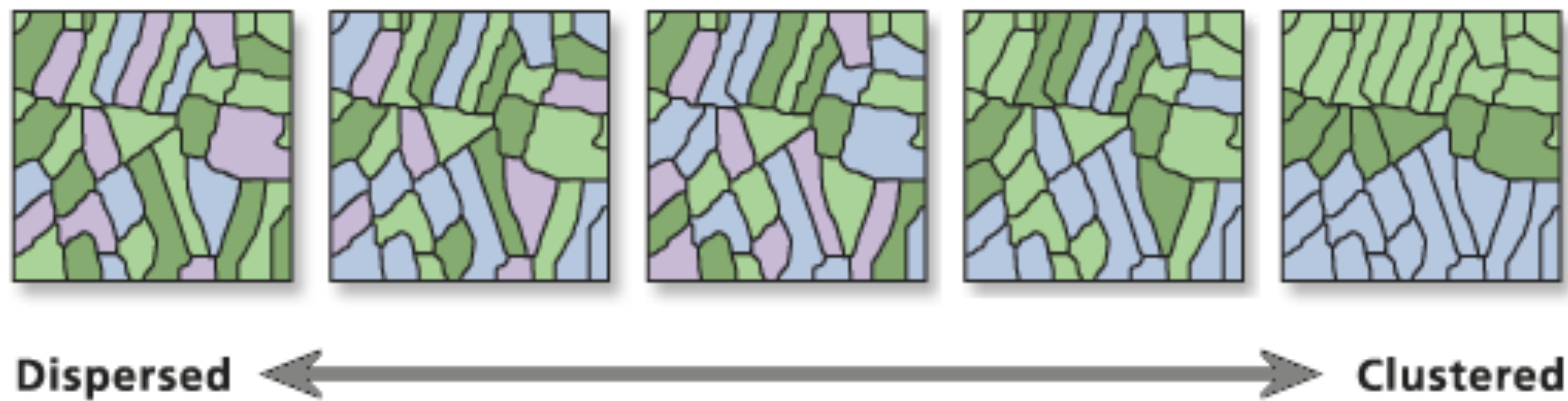
```
from splot.libpysal import plot_spatial_weights
```

```
plot_spatial_weights(w_queen, db, indexed_on="area_id")
```



Jupyter

Next week: Spatial autocorrelation



Geospatial Data Science Exam Project Instructions, Spring 2026

Sources and further materials for today's class



***Geographic Data Science
with Python***



https://geographicdata.science/book/notebooks/05_choropleth.html

https://darribas.org/gds_course/content/bE/concepts_E.html

[https://geodacenter.github.io/lab/?lab=./workbook/4c_distance_functions/lab4c.html&title=Spatial Weights as Distance Functions](https://geodacenter.github.io/lab/?lab=./workbook/4c_distance_functions/lab4c.html&title=Spatial%20Weights%20as%20Distance%20Functions)